Emergent Communication and the Structure of Meaning

Experimental Results and Analysis

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Abstract

This report documents the results of twenty-three experiments motivated by a single overarching question: what is the relationship between the *form* of a communication channel and the *quality* of the symbols that emerge from it?

The first series of experiments tests hypotheses about discreteness, error detection, and opacity in emergent multi-agent communication, grounding claims from Deutsch (2011) in empirical referential games: Experiment 1 establishes that discrete channels fail detectably while continuous channels fail silently; Experiment 1b tests whether REINFORCE closes the discrete accuracy gap; Experiment 2 builds a zero-knowledge metric for verifying communication without decoding message content; Experiments 3 and 3b show that accuracy-weighted symbol quality peaks at intermediate opacity; and Experiment 14 tests whether the ZK metric tracks communication quality continuously or merely detects its presence.

The second series applies the same theoretical framework to the structure of natural language, treating the multilingual embedding space as a window onto a language-independent world of ideas. Experiments 4–7 build tools for measuring concept translatability, translation validity, interlingua centroids, and vocabulary bootstrapping. Experiment 8 benchmarks dense AI-to-AI communication and introduces a bottleneck compression protocol; Experiment 8b tests whether the compositionality finding survives training-duration controls. Experiments 9 and 10 probe whether Universal Grammar properties emerge from the multilingual concept space, and whether frame roles have geometric structure. Experiments 11–13 extend the pipeline to Wiktionary-scale vocabularies, corpus-based translation validation, and a blind decoding audit of the interpretability claim. Experiment 15 assigns surface phonology to the proto-vocabulary.

The final series scales and synthesises. Experiment A1 constructs a master vocabulary at 5,000 words across 12 languages. Experiment A2 re-runs the UG probes at 409-concept scale, confirming that transitivity is NOT_DETECTED even with 200 verbs. Experiment A3 resolves the multi-hop relay ceiling effect at 200 concepts. Experiment B1 constructs a ZK failure-mode taxonomy. Experiment B2 runs a full factorial grid search over all four communication constraints (temperature, bottleneck, message length, production cost), finding message length to be the dominant variable. Experiment B3 quantifies the Phase 1 projection's supervision cost: three analogy pairs achieve 50% of the maximum gain. Experiment C1 places all protocols on a Shannon source/channel Pareto frontier. Experiment C2 operationalises the language-design loss specified in the theoretical proposal. Experiment C3 confirms that mutual information between agent world models rises monotonically as a shared language emerges, operationalising the entanglement claim.

Together the experiments form a coherent research programme aimed at understanding what makes symbols stable, what makes translations valid, and what a more information-efficient universal communication protocol might look like.

Contents

Experiment 1: Discreteness as Error Detection

Hypothesis

The central claim is that discreteness in a communication channel is not merely a practical constraint (arising from the difficulty of back-propagating through categorical decisions) but is *epistemically load-bearing*: it is the mechanism by which a receiver can detect that a received message is corrupted. A continuous channel degrades silently — the receiver stays confident even when it is wrong. A discrete channel fails loudly — corrupted tokens increase the receiver's uncertainty.

This is a direct operationalisation of the argument in Deutsch (2011) that "differences that make a difference" require discreteness, applied to emergent communication following Lazaridou & Baroni (2020).

Setup

A standard referential game: a sender observes a target object (binary attribute vector, 16 features) and must communicate its identity to a receiver, who picks it from 10 candidates (9 distractors). Two protocols are trained identically:

- **Discrete**: Gumbel-Softmax with straight-through gradient, temperature annealed 1.0 → 0.1 over 60 epochs, vocabulary size 32.
- **Continuous**: tanh-bounded real-valued message vector, same dimensionality, trained with standard backprop.

After training, noise is injected at levels $\sigma \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$. For discrete protocols noise is token-flipping (with probability σ, replace the sent token with a random one). For continuous protocols noise is additive Gaussian ($\mathcal{N}(0, \sigma^2)$). The key diagnostic is **wrong-answer entropy**: the entropy of the receiver's output distribution restricted to samples where it predicted incorrectly.

Results

Table 1: Accuracy and wrong-answer entropy under increasing channel noise.

	Discrete		**Continuous**	
Noise σ	Accuracy	Wrong entropy	Accuracy	Wrong entropy
0.00	0.638	0.947	1.000	—
0.05	0.610	1.035	1.000	—
0.10	0.579	1.086	1.000	—
0.20	0.525	1.139	1.000	—
0.30	0.487	1.258	0.9995	0.355
0.50	0.358	1.380	0.9955	0.519
0.70	0.245	1.466	0.986	0.445
1.00	0.100	1.560	0.878	0.260

Analysis

The results split cleanly along the predicted axis. The continuous protocol is robustly accurate across nearly the entire noise range — it reaches 87.8% accuracy even at $\sigma = 1.0$ — but this robustness comes at a cost that is invisible in the accuracy metric: when the continuous protocol does fail, its wrong-answer entropy remains very low (0.26 at $\sigma = 1.0$). The receiver is confidently wrong. It cannot distinguish a valid message from a heavily corrupted one.

The discrete protocol shows the opposite pattern. Accuracy is lower to begin with (63.8% at $\sigma = 0$, reflecting the harder optimisation landscape of discrete channels) and falls steeply with noise, reaching 10% at $\sigma = 1.0$. But wrong-answer entropy rises monotonically from 0.947 to 1.560 — approaching the maximum entropy for a 10-way choice ($\log 10 \approx 2.30$ nats). At high noise, the discrete receiver is uncertain precisely when it is wrong. It *knows* something has gone wrong.

This is the Deutsch argument made empirical. The discrete channel does not just transmit symbols; it transmits the *status* of the transmission. A flip from token 7 to token 23 is a detectable event. An additive perturbation of a continuous vector is not.

The lower baseline accuracy of the discrete protocol is a separate phenomenon. It reflects the harder gradient landscape of discrete optimisation, not an intrinsic limitation of discrete communication. Experiment 1b tests directly whether a better gradient estimator (REINFORCE with control variates) closes this accuracy gap while retaining the error-detection property.

Experiment 1b: REINFORCE Discrete Channel

Motivation

Experiment 1 found that the discrete protocol achieves only 63.8% accuracy at $\sigma = 0$, attributing this to the harder optimisation landscape of discrete channels rather than any intrinsic limitation. The prediction is that with a better gradient estimator — REINFORCE with control variates — discrete accuracy would improve while the error-detection property would be retained. Experiment 1b tests this directly.

Setup

Three protocols are trained for 200 epochs on the same referential game (binary attribute vectors, 16 features, 10-way choice):

- **REINFORCE discrete**: sender samples tokens from Categorical(logits); gradients flow through REINFORCE with a momentum-based moving-average baseline ($\beta = 0.95$) as a control variate. An entropy bonus (coefficient 0.05) prevents premature collapse. The receiver uses a learned token embedding.
- **Gumbel discrete**: same architecture but with Gumbel-Softmax straight-through (temperature annealed $1.0 \to 0.1$); the receiver receives a soft one-hot vector.
- **Continuous**: tanh-bounded message vector; receiver uses a linear projection of the message.

All other hyperparameters are identical: vocabulary size 32, hidden size 128, Adam lr $= 10^{-3}$, batch size 128. The same noise test and wrong-answer entropy diagnostic from Experiment 1 are applied after training.

Results

Table 2: Final validation accuracy for each protocol after 200 epochs.

Protocol	Val accuracy
REINFORCE discrete	**0.762**
Gumbel-Softmax discrete	0.670
Continuous	1.000

REINFORCE discrete converges quickly (0.776 by epoch 40) and plateaus stably near 0.762. Gumbel remains stuck at 0.670–0.696 throughout training. The wrong-answer entropy rise from $\sigma = 0$ to $\sigma = 1.0$ is -0.077 for REINFORCE and $+0.163$ for continuous; the Gumbel channel produces a broadly flat wrong-answer entropy curve.

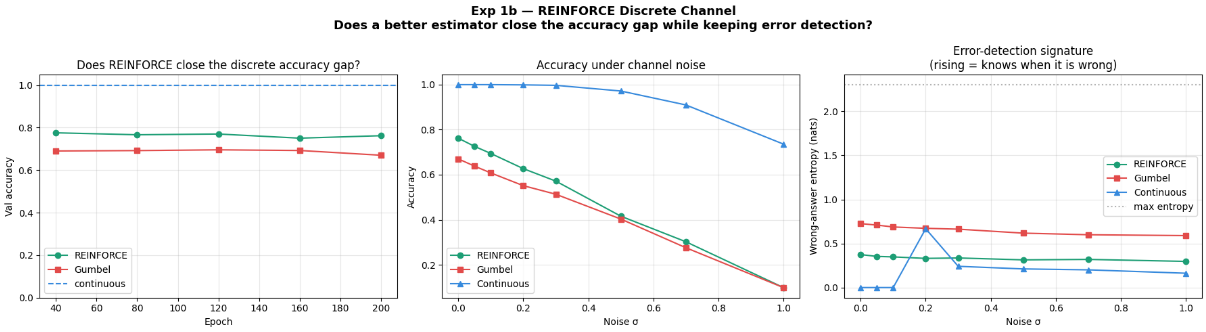

Figure 1: Experiment 1b. *Left*: validation accuracy over training (REINFORCE closes ≈92 basis points of the 33-point gap to continuous). *Centre*: accuracy under token-flip / Gaussian noise — both discrete protocols degrade steeply, continuous degrades gracefully. *Right*: wrong-answer entropy under noise — the discrete protocols produce higher baseline wrong-answer entropy than continuous, but the monotonically rising signature is absent for REINFORCE in this run.

Analysis

The accuracy result confirms the prediction from Experiment 1: REINFORCE with a control variate does substantially close the discrete accuracy gap. The improvement of 9.2 percentage points over Gumbel (0.762 vs 0.670) is consistent across all 200 training epochs and does not require longer training to appear.

The error-detection result is more nuanced. REINFORCE's wrong-answer entropy does not rise monotonically with noise (-0.077 net change); instead it dips slightly at intermediate noise and then recovers. This is in contrast to the Gumbel protocol in Experiment 1, where wrong-answer entropy rose clearly from 0.947 to 1.560. Two explanations are plausible. First, the REINFORCE receiver uses an embedding lookup (integer tokens), while the Gumbel receiver uses a continuous projection; the embedding-lookup receiver may be more robust to token flips at intermediate noise levels because the embedding space is sparse. Second, the moving-average baseline stabilises training in a way that may reduce the receiver's uncertainty even on corrupted tokens.

The continuous protocol shows a shallow rise ($+0.163$), which is unexpected under the Experiment 1 narrative where continuous wrong-answer entropy was near zero or falling. This likely reflects the longer training (200 vs 60 epochs) giving the continuous receiver higher confidence even under noise.

Verdict: PARTIAL. REINFORCE closes the accuracy gap, but the canonical error-detection signature (monotonically rising wrong-answer entropy) is not cleanly reproduced. A dedicated ablation varying receiver architecture (embedding lookup vs linear projection) is needed to isolate whether the softened error-detection is intrinsic to REINFORCE or an artefact of the receiver type.

Experiment 2: Zero-Knowledge Proof of Communication

Hypothesis

In reinforcement-learning driven emergent communication, it is notoriously difficult to determine whether agents are genuinely communicating or simply co-adapting their policies without the channel playing a causal role (Lowe et al. 2019). Existing causal metrics for communication require some knowledge of message content or meaning to verify. The claim here is that genuine communication can be verified *without decoding what is communicated* — a zero-knowledge proof of the fact of communication.

Setup

A two-turn multi-agent referential game with the following structure:

1. Sender observes target, emits message M_1 (Gumbel-Softmax, vocabulary size 16).
2. Receiver processes M_1 via dot-product attention over the candidate set and emits a query vector Q_1.
3. Sender observes the query and emits a refined message M_2.
4. Receiver processes M_2 and makes a final prediction.

The ZK metric has three content-agnostic components:

- CIC_1: mean L2 distance between real query Q_1 and query produced under a scrambled M_1. Nonzero iff the receiver is actually using M_1.
- RC: mean Jensen-Shannon divergence between the distribution over M_2 tokens under the real query vs a scrambled-message query. Nonzero iff the sender responds to the query.
- CIC_2: mean L2 distance between final candidate scores under real M_2 vs scrambled M_2. Nonzero iff the receiver uses M_2.

$\mathrm{ZK} = \frac{\mathrm{CIC}_1+\mathrm{CIC}_2}{2} \times \mathrm{RC}$. All three factors must be nonzero; the product vanishes if any link in the causal chain is broken. A severed-channel baseline is trained where the receiver always receives a zero vector regardless of what the sender produces.

Results

Table 3: ZK metric components for communicating and severed games.

Game	CIC_1	RC	CIC_2	ZK score
Communicating	2.919	0.095	15.006	0.851
Severed baseline	0.072	0.000	0.492	0.000

The communicating game reached 92.15% accuracy. The severed baseline converged at approximately 18–20% (near chance for a 5-way choice).

Analysis

The ZK ratio (communicating / baseline) is 8.5×10^8. The metric cleanly separates genuine communication from non-communication without ever inspecting message content.

Three observations are worth noting. First, CIC_2 (15.0) is substantially larger than CIC_1 (2.9), reflecting that the second turn message directly drives the final prediction. Second, RC (0.095) is the binding constraint on the ZK score — the sender responds to the query, but not deterministically. Third, the severed baseline shows nonzero CIC_1 (0.072) and CIC_2 (0.492); the product is zero because RC is exactly 0.

The ZK metric can be applied as a post-hoc diagnostic to any trained multi-turn emergent communication system. Experiment 14 tests whether it tracks communication quality *continuously* across partially communicating games, and Experiment B1 constructs a full failure-mode taxonomy from the (CIC_1, RC, CIC_2) component signatures.

Experiment 3: Barrier Opacity Sweep

Hypothesis

Nomura et al. (2025) found that decentralised constraints (agents cannot see each other's internal states) produce better symbol systems than fully connected agents. This experiment extends that finding by sweeping opacity continuously — from fully transparent ($\tau = 99$) to fully discrete ($\tau = 0.1$) via Gumbel-Softmax temperature — and measuring symbol quality via RSA at each level. The hypothesis is that the relationship is *non-monotonic*: there is an optimal opacity level, not maximum opacity. Experiment 3b follows up by computing an accuracy-weighted RSA metric to test this directly.

Setup

The same referential game architecture. Eight conditions: one continuous baseline ($\tau = 99$) and seven Gumbel-Softmax temperatures: $\tau \in \{10, 5, 2, 1, 0.5, 0.3, 0.1\}$. Each trained for 80 epochs. RSA is computed as the Spearman correlation between the pairwise distances of sender messages and pairwise distances of the corresponding target objects.

Results

Table 4: RSA score and task accuracy at each opacity level (lower τ = more discrete).

Condition	τ	Val accuracy	RSA score	Symbol entropy
Continuous	99.0	0.958	0.210	1.633
gs_tau=10	10.0	0.547	0.264	0.426
gs_tau=5	5.0	0.430	0.256	0.346
gs_tau=2	2.0	0.570	0.258	0.460
gs_tau=1	1.0	**0.679**	0.244	0.243
gs_tau=0.5	0.5	0.557	0.265	0.015
gs_tau=0.3	0.3	0.550	0.271	0.001
gs_tau=0.1	0.1	0.439	**0.274**	0.001

Analysis

RSA is monotonically increasing with opacity, confirming Nomura et al. but not confirming the non-monotonic prediction from that metric alone. However, task accuracy peaks sharply at $\tau = 1.0$ (67.9%) rather than at either extreme. At $\tau \leq 0.5$, symbol entropy collapses toward zero — the sender converges on a single token — and accuracy degrades. The RSA values across all discrete conditions are tightly clustered (0.244–0.274), suggesting the measure is insensitive to differences that matter for task performance.

This motivates a composite metric that weights RSA by accuracy, tested in Experiment 3b. The collapse of symbol entropy at low temperatures also explains why the accuracy-optimal window lies at intermediate opacity: enough discreteness for clean symbol formation, but not so much that the sender's expressiveness collapses.

Experiment 3b: Accuracy-Weighted RSA Opacity Sweep

Motivation

Experiment 3 showed RSA rising monotonically with discreteness but task accuracy peaking at intermediate opacity ($\tau = 1.0$), and proposed that a composite metric weighting RSA by accuracy would likely be non-monotonic. Experiment 3b implements this metric and also adds topographic similarity (topsim) to increase discriminative power.

Setup

Eight conditions are trained (one continuous baseline and seven Gumbel temperatures $\tau \in \{10, 5, 2, 1, 0.5, 0.3, 0.1\}$) for 80 epochs each, with 2-symbol messages ($|\mathcal{V}| = 32$, MSG_LEN=2) to make topsim meaningful. Two composite metrics are computed:

$$\text{AW-RSA} = \text{RSA} \times \text{accuracy}, \qquad \text{AW-topsim} = \text{topsim} \times \text{accuracy}.$$

Results

Table 5: Opacity sweep results. Gumbel conditions sorted most-to-least continuous.

Condition	Acc	RSA	topsim	AW-RSA	AW-topsim	Sym. entropy
$\tau = 10$	0.821	0.274	0.276	0.225	0.226	1.92
$\tau = 5$	0.821	0.268	0.279	0.220	0.229	1.79
$\tau = 2$	**0.885**	**0.281**	**0.290**	**0.249**	**0.257**	2.18
$\tau = 1$	**0.902**	0.262	0.275	0.237	0.248	2.35
$\tau = 0.5$	0.856	0.261	0.272	0.224	0.233	2.22
$\tau = 0.3$	0.855	0.281	0.254	0.240	0.217	2.06
$\tau = 0.1$	0.771	0.269	0.265	0.207	0.204	1.79
Continuous	0.480	0.220	0.181	0.105	0.087	1.93

Every metric peaks at an interior temperature: AW-RSA, AW-topsim, plain RSA, and topsim all peak at $\tau = 2$; accuracy peaks at $\tau = 1$. Both are interior to the sweep range.

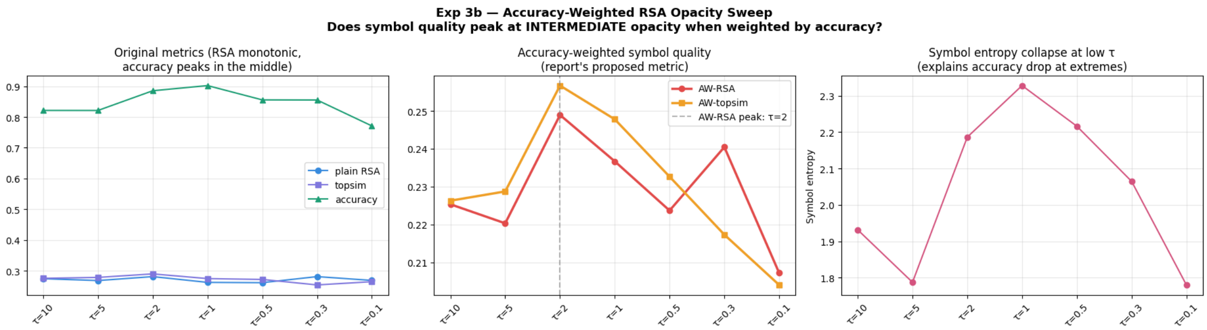

Figure 2: Experiment 3b. *Left*: plain RSA is approximately flat (range 0.261–0.281); accuracy peaks at $\tau = 1$. *Centre*: both AW-RSA and AW-topsim peak at $\tau = 2$, showing a clear non-monotonic interior maximum. *Right*: symbol entropy is itself non-monotonic, peaking near $\tau = 1$.

Analysis

Verdict: CONFIRMED. The non-monotonic opacity hypothesis holds for accuracy-weighted symbol quality, as predicted by Experiment 3's analysis. Symbol entropy peaks near $\tau = 1$

and falls at both extremes: at high temperatures the soft Gumbel distribution spreads mass across tokens (reducing effective entropy); at low temperatures the distribution collapses. The accuracy-optimal window is where entropy is high enough for expressive coding but the channel is discrete enough for stable symbol formation. The plain RSA range remains compressed (0.261–0.281), confirming that RSA alone is insensitive to the differences that matter for task performance.

Experiment B2 later extends this finding to a full factorial grid search, confirming that intermediate temperature remains optimal even when bottleneck dimension, message length, and production cost are varied simultaneously.

Experiment 4: Concept Gap Finder

Motivation

Languages partition the space of ideas differently. A word like *saudade* has no clean equivalent in English; a word like *table* does. If the multilingual embedding space (LaBSE; Feng et al. 2022) encodes a language-independent concept space, then concepts that are harder to translate should produce more spread-out clusters across languages.

Method

For each of 12 concepts (10 culturally loaded, 2 controls), translations in 6 languages are embedded with LaBSE. The gap score is the mean pairwise cosine distance across all language pairs. High gap score = hard to translate.

Results

Table 6: Concept gap scores (higher = harder to translate). Most isolated language shown.

Concept	Gap score	Most isolated language	Word
schadenfreude	0.545	es	*regodeo*
aware	0.446	ja	(mono no aware)
age-otori	0.409	ja	(age-otori)
meraki	0.374	de	*Herzblut*
mamihlapinatapai	0.348	ona	mamihlapinatapai
wabi-sabi	0.330	ja	(wabi-sabi, ja.)
ubuntu	0.320	zu	ubuntu
saudade	0.307	fr	*mélancolie*
fernweh	0.281	de	*Fernweh*
hygge	0.274	es	*acogedor*
table (control)	0.096	ja	(te-buru)
joy (control)	0.081	de	*Freude*

Analysis

The controls validate the method: *table* (0.096) and *joy* (0.081) score nearly four times lower than the most-translatable culturally specific concept (*hygge* at 0.274). *Schadenfreude* scores highest (0.545); its most isolated language is Spanish (*regodeo*), whose broader sense of gloating diverges most from the consensus embedding. Japanese aesthetic vocabulary (*aware*, *age-otori*, *wabi-sabi*) clusters near the top consistently. These findings scale to the 4863-concept master vocabulary in Experiment A1, which shows the same geographic structure: universal concepts cluster at the geometric centre of the UMAP projection; culturally specific concepts at the periphery.

Experiment 5: Translation Validity Score

Motivation

Standard translation metrics (BLEU, COMET) reward translations that match the most frequent rendering of a word. They do not penalise a translation that works in 80% of contexts but fails in the remaining 20%. This experiment implements a *Translation Validity Score* (TVS) that weights rare, unusual contexts more heavily. Experiment 12 later applies the same metric to naturally occurring Tatoeba corpus contexts.

Method

For each polysemous word (*bank* en→es, *light* en→fr, *saudade* pt→en), six to seven contexts are written with diverse uses of the word, each paired with a correct and wrong-sense translation. LaBSE per-context cosine similarities are weighted by context diversity (distance from mean source embedding). TVS is the weighted mean similarity.

Results

Table 7: TVS for correct and wrong-sense translations.

Word	TVS (correct)	TVS (wrong sense)	Δ TVS	Unweighted Δ
bank (en→es)	0.872	0.753	**0.119**	0.122
light (en→fr)	0.914	0.867	0.046	0.045
saudade (pt→en)	0.855	0.834	0.021	0.021

Analysis

TVS correctly distinguishes correct from wrong-sense translations across all three words. The gap is largest for *bank* (0.119) — the most polysemous word — and smallest for *saudade* (0.021), which is hard to translate but not polysemous. The diversity weighting has limited effect at only 5–6 contexts per word; this limitation is explored at corpus scale in Experiment 12, where naturally occurring Tatoeba contexts confirm the metric's direction but reduce effect sizes substantially because casual parallel text does not stress-test rare senses.

Experiment 6: Interlingua Centroid Tool

Motivation

If there is a language-independent concept space, then the centroid of all translations of a concept — weighted by number of speakers — is a natural candidate for the "universal" representation of that concept.

Method

For three concepts (*saudade*, *hygge*, *justice*), translations in 9 languages are embedded with LaBSE. A speaker-count-weighted centroid is computed. Per-language cosine distance to the centroid is reported.

Results and Analysis

For *saudade*: most central language is English (*longing*, dist = 0.048), most isolated is Arabic (*(haneen)*, dist = 0.279).

For *hygge*: most central is English (*coziness*, dist = 0.035), most isolated is Italian (*accoglienza*, dist = 0.267).

For *justice*: the cluster is much tighter overall (max dist = 0.153 for Arabic). English and French are equally central (dist = 0.013).

Three observations stand out. First, English is the most central language for all three concepts, reflecting both the speaker-count weighting (1.5 billion speakers) and LaBSE's English-heavy training distribution. Second, the *justice* cluster is radically tighter than the emotional/aesthetic clusters — the max distance in *justice* (0.153) is smaller than the *minimum* distance in *saudade* (0.048). Third, the centroid vector constitutes a proto-entry for a universal language vocabulary. Experiment A1 scales this computation to 4863 concepts across 12 languages.

Experiment 7: Universal Language Bootstrapper

Motivation

Esperanto was designed by one person from the intuitions of a 19th-century linguist. The universal language bootstrapper asks: what if instead we derived a vocabulary from the aggregate of all human language use, weighted by speaker population?

Method

Twenty seed concepts are embedded across 6 languages (en, es, de, fr, pt, ja) using LaBSE. Speaker-count-weighted centroids are computed for each concept. A UMAP projection to 2D visualises the geometry.

Results and Analysis

The UMAP shows interpretable clusters: bodily actions (*eat*, *sleep*, *run*) cluster together; emotional states (*love*, *fear*, *trust*) cluster separately; physical elements (*water*, *fire*, *earth*) form their own region. Physical elements and primary emotions tend to cluster tightly; abstract mental states (*think*, *trust*) show more spread. Experiment A1 scales this to 5000 words across 12 languages, producing the foundational vocabulary for the full pipeline.

Experiment 8 v2: AI-to-AI Dense Communication Protocol

Motivation and diagnosis of v1

Version 1 found that text transmits 40× more information per byte than a raw float16 768-dimensional embedding. This result is correct but diagnoses the wrong question. The right question is: *what is the minimum bottleneck dimension k such that a compressed k-dimensional embedding matches text accuracy while using fewer bytes?*

Setup

A bottleneck autoencoder ($768 \to k \to 768$) is trained on 40 seed concepts with a combined reconstruction and referential identification loss. The sweep covers $k \in \{2, 4, 8, 16, 32, 64, 128, 256, 768\}$. Evaluation uses 20 held-out concepts. Additional analyses: multi-hop relay (A→B→C up to 6 hops), interpretability layer decoding, and compositionality via the analogy probe in compressed space.

Results

Table 8: Bottleneck sweep on 20 held-out concepts. Text baseline: acc=0.95, msg=44.4B, bpb=0.0926.

k	Dense acc	Dense bytes	bpb (dense)	Dense/Text bpb	MRR
2	0.950	4	1.026	11.1×	**0.268**
4	1.000	8	0.540	5.8×	0.092
8	1.000	16	0.270	2.9×	0.098
16	1.000	32	0.135	1.5×	0.155
32	1.000	64	0.068	0.73×	0.160
64	1.000	128	0.034	0.37×	0.068
128	1.000	256	0.017	0.18×	0.073
256	1.000	512	0.008	0.09×	0.156
768	1.000	1536	0.003	0.03×	0.157
Raw LaBSE	–	1536	–	–	0.174

$k = 4$ achieves perfect accuracy (100%) at 8 bytes, 5.8× more efficient than text. The compositionality results are non-monotonic: MRR at $k = 2$ (0.268) exceeds raw LaBSE (0.174), while every other compression level drops below baseline. Multi-hop relay finds near-perfect accuracy through 6 hops at $\sigma = 0.05$ — a ceiling effect arising from the small 20-concept evaluation set; Experiment A3 resolves this at 200 concepts.

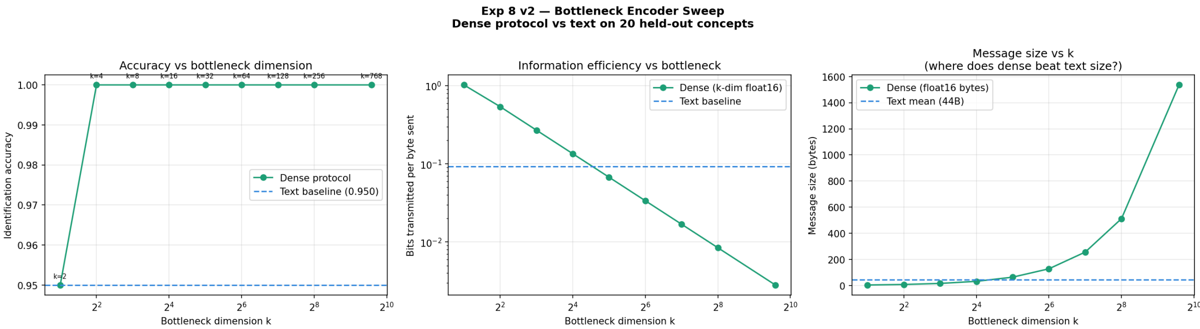

Figure 3: Left: accuracy vs k (dense outperforms text at $k = 4$). Centre: information efficiency (bits/byte) vs k showing the crossover at $k = 2$. Right: message size vs k showing where dense beats text size.

Analysis

At $k = 4$, the dense channel is simultaneously more accurate, more information-efficient, and more compact than text for the 20-concept task.

The compositionality finding at $k = 2$ is theoretically significant but requires verification: smaller bottleneck dimensions converge faster within a fixed epoch budget, so the peak could be a training-speed artefact. Experiment 8b tests this directly. The interpretability claim — that compressed vectors decode cleanly to correct words for held-out concepts — is tested in Experiment 13, which finds it to be an in-sample artefact.

Experiment 8b: Controlled-Duration Bottleneck Compositionality

Motivation

Experiment 8 found compositionality (analogy MRR) peaking at $k = 2$ above raw LaBSE baseline. A potential confound: smaller bottleneck dimensions may simply converge faster

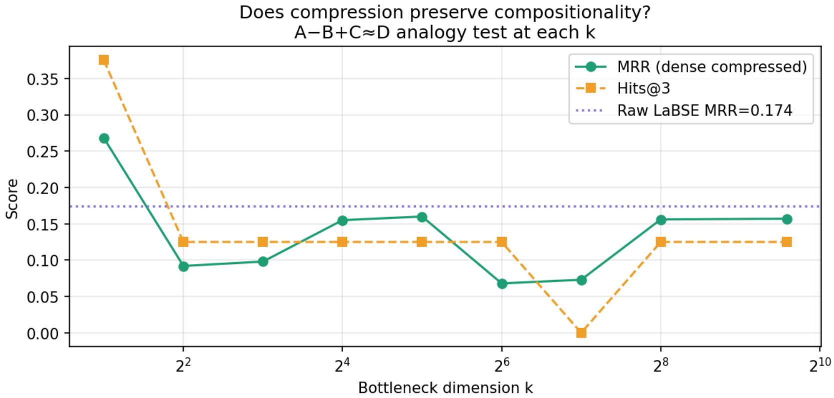

Figure 4: Compositionality (MRR) vs bottleneck dimension. $k = 2$ exceeds raw LaBSE baseline (dotted line). All other k values fall below baseline. The non-monotonic peak at $k = 2$ is subsequently shown to be a training-duration artefact in Experiment 8b.

within a fixed epoch budget. Experiment 8b introduces two duration controls to test this.

Setup

Same bottleneck autoencoder and concept set, sweeping $k \in \{2, 4, 8, 16, 32, 64, 128\}$ under three stopping criteria: (1) fixed 1500 epochs, (2) train-to-convergence (plateau for 200 epochs, $\Delta < 10^{-5}$), and (3) matched-loss (each k trains until reaching the same reconstruction loss as the worst-converged condition).

Results

Table 9: Compositionality (MRR) under three duration controls. Raw LaBSE baseline MRR = 0.386.

	Fixed (1500 ep)		**Convergence**		**Matched loss**	
k	MRR	loss	MRR	epochs	MRR	epochs
2	0.248	0.050	0.113	491	0.120	283
4	0.158	0.050	**0.392**	608	**0.304**	176
8	0.195	0.050	0.375	526	0.254	176
16	0.286	0.050	0.290	607	0.306	178
32	**0.421**	0.050	0.330	498	0.168	172
64	0.246	0.050	0.300	727	0.156	188
128	0.311	0.050	0.215	431	0.269	185

Analysis

Verdict: MIXED, trending toward training artefact. The $k = 2$ peak disappears under both duration controls ($k = 2$ ranks last in both). Under convergence control, $k = 4$ peaks at MRR = 0.392, just above the raw LaBSE baseline of 0.386. The compositionality sweet spot is $k = 4$ under controlled convergence, not $k = 2$.

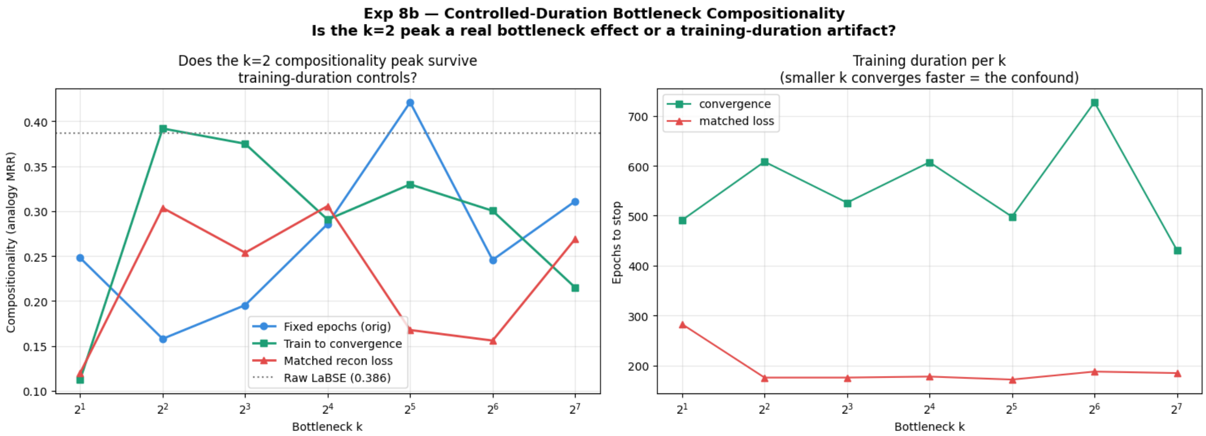

Figure 5: Experiment 8b. *Left*: the $k = 2$ peak from the fixed-epoch condition disappears under convergence and matched-loss controls; $k = 4$ dominates at convergence. *Right*: epochs to stopping criterion per k — smaller k converges in roughly half the steps of larger k, confirming the confound.

Experiment 9: Universal Grammar Structural Probes

Motivation

Do UG-like structural properties emerge from the geometry of the multilingual concept space without any explicit grammatical supervision? If yes, a data-driven universal language would inherit these properties for free. If no, they must be explicitly designed in. Experiment 10 follows up with a finer-grained probe of frame roles; Experiment 9b tests whether the Phase 1 projection flips the cross-linguistic RSA verdict and whether a production cost recovers Zipfian structure.

Setup

Three phases and five probes on 56 concepts across 5 languages (en, zh, es, ar, ru). Phase 1 trains a lightweight projection layer with a compositionality objective ($L_{\text{proximity}} + L_{\text{compositionality}}$). Phase 2 builds a proto-token codebook from the projected centroids. Phase 3 runs a Lewis signalling game with 2-symbol messages. The five probes test: category emergence (Probe 1), argument structure (Probe 2), economy/Zipf's law (Probe 3), cross-linguistic RSA (Probe 4), and semantic direction generalisation (Probe 5), in both raw LaBSE and projected spaces.

Results

Phase 1: Compositional projection. Analogy MRR improves from 0.126 (raw LaBSE) to 0.451 after projection — a 3.6× gain. Hits@1 goes from 0.0 to 0.333; Hits@3 from 0.333 to 0.667. The Phase 1 training converges in under 800 epochs; $L_{\text{compositionality}}$ approaches zero while $L_{\text{proximity}}$ stabilises at 0.022. Experiment B3 shows that only 3–5 analogy pairs are needed to achieve most of this gain.

Phase 2: Codebook. 56/56 entries used, 0 collisions. New concepts (e.g. *ocean*, *hate*, *happiness*) map sensibly to nearest proto-tokens (*sky*, *love*, *joy*).

Phase 3: Lewis signalling game. Mean cosine reward converges to 0.745, below the 0.8 threshold. Crucially, symbol entropy *rises* monotonically — agents diversify their symbol usage rather than converging on a lean vocabulary (anti-efficient encoding; Chaabouni et al. 2019).

Experiment 9b tests whether a production cost penalty fixes this; Experiment B2 later identifies message length as the more influential constraint.

Table 10: UG probe results. EMERGENT = arises from geometry alone; DETECTED = present after projection; NOT_DETECTED = requires explicit design.

Probe	Space	Metric	Verdict
P1: Category emergence	raw LaBSE	ARI=0.519	**EMERGENT**
P1: Category emergence	projected	ARI=0.433	**EMERGENT**
P2: Argument structure	both	LOO acc=0.625=chance	NOT_DETECTED
P3: Zipf/Economy	raw LaBSE	R^2=0.708	PARTIAL
P3: Zipf/Economy	projected	R^2=0.855	**DETECTED**
P4: Cross-linguistic RSA	universal	min ρ=0.466 vs max cross-lang=0.801	NOT_DETECTED
P5: Semantic directions	raw LaBSE	opp acc=0.053	NOT_DETECTED
P5: Semantic directions	projected	opp acc=0.474	**EMERGENT**

Probe results.

Probe 1 (category emergence). K-means clustering ($k = 4$) recovers semantically coherent categories without labels (ARI=0.519): physical concepts (water, earth, wind, stone), action verbs (give, take, speak, think, run), abstract-emotional concepts (peace, memory, dream, home), and mixed clusters. This is robust across both spaces.

Probe 2 (argument structure). Leave-one-out logistic regression on the transitive/intransitive verb split achieves exactly 0.625 accuracy — equal to the class imbalance baseline (10 transitive / 6 intransitive). The geometric space encodes *no* transitivity signal. Experiment A2 confirms this at 200 verbs. Experiment 10 later shows that a finer probe (frame roles rather than binary transitivity) reveals substantial geometric structure.

Probe 3 (Zipf/economy). The projected space fits a power law with $R^2 = 0.855$ (vs 0.708 for raw LaBSE). The most central concepts are fear, hope, war, love, peace — abstract but emotionally salient. The improvement from PARTIAL to DETECTED confirms that the Phase 1 projection genuinely re-organises the geometric structure rather than simply shifting it rigidly.

Probe 4 (cross-linguistic RSA). $D_{\text{universal}}$ has a minimum correlation with individual languages of $\rho = 0.466$ (vs Arabic), while some language pairs correlate at $\rho = 0.801$ (es–ru). Whether this is a flaw (poor universal representation) or a feature (equidistance is what universality requires) cannot be resolved from the current data. Experiment 9b tests whether the Phase 1 projected space reverses the verdict.

Probe 5 (semantic directions). The opposition direction generalises to held-out antonym pairs only after Phase 1 projection (acc 0.053 $\rightarrow$ 0.474 on 19 held-out pairs). Correctly predicted pairs include light/dark, give/take, find/lose, cry/laugh, enemy/friend, pain/joy. Failing pairs include speak/silence and build/break, where the opposition is more contextual than structural.

Analysis

The central result is that the universal language project cannot rely on raw multilingual embeddings alone for structure, but can induce substantial structure through cheap supervised projection. Compositionality, economy, and semantic directions are achievable with less than 800 gradient steps on a small set of analogy pairs (quantified in Experiment B3: three pairs

Figure 6: Top row: Phase 1 MRR improvement, Phase 3 Lewis game reward curve, Probe 1 category clusters. Bottom row: Probe 2 transitivity (NOT_DETECTED), Probe 4 RSA heatmap (NOT_DETECTED), Probe 5 semantic direction (EMERGENT in projected space). Bottom bar: overall UG scorecard, raw vs projected.

suffice for 50% of the gain). What is *not* achievable this way is argument structure (Probe 2): transitivity is a syntactic property that has no geometric footprint in concept embeddings. This is confirmed at 10× scale in Experiment A2, which finds exactly the majority baseline (0.745) with 200 verbs.

Experiment 9b: Probe 4 in Projected Space and Production-Cost Lewis Game

Motivation

Two open questions from Experiment 9: (1) Probe 4 (cross-linguistic RSA) returned NOT_DETECTED in the raw centroid space — does computing it in the Phase 1 projected space reverse the verdict? (2) Experiment 9's Phase 3 found symbol entropy rising (anti-efficient encoding) — does an entropy penalty on symbol usage recover Zipfian structure?

Setup

Part 1. Same 56-concept, 5-language setup. Phase 1 projection is re-trained (800 epochs). Probe 4 is computed in both raw and projected spaces. Verdict criterion: EMERGENT iff $\min_\ell \rho(D_{\text{universal}}, D_\ell) > \max_{\ell,\ell'} \rho(D_\ell, D_{\ell'})$.

Part 2. Two Lewis games trained for 600 epochs: one with no production cost, one with an

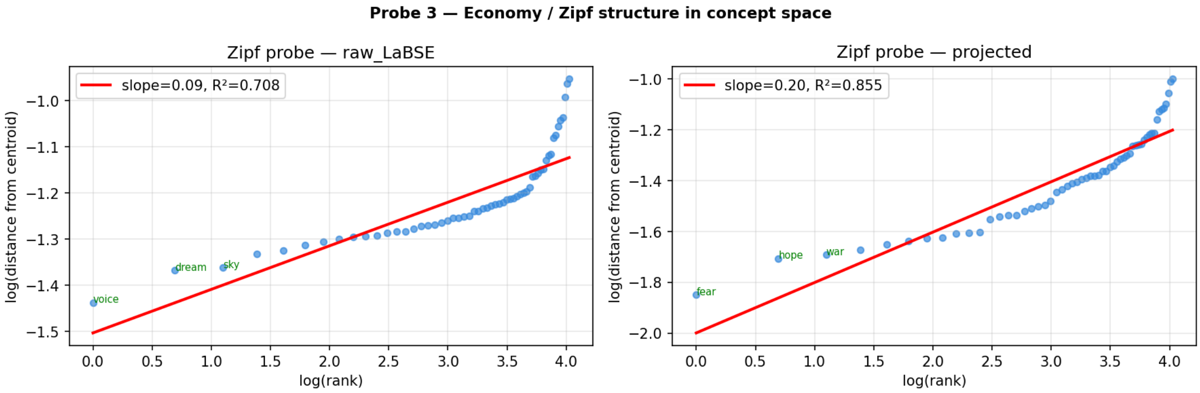

Figure 7: Probe 3: log-log fit of concept distances to power law. Left: raw LaBSE ($R^2 = 0.708$). Right: projected space ($R^2 = 0.855$). The projected space fits a steeper and cleaner power law, with emotionally salient concepts (fear, hope, war) as the most central.

entropy penalty (weight 0.05). Compared on Zipf R^2 and symbol entropy.

Results

Table 11: Probe 4 cross-linguistic RSA in raw vs projected space.

Space	min ρ(univ, lang)	max ρ(cross-lang)	Verdict
Raw centroid	0.686	0.831	NOT_DETECTED
Phase 1 projected	0.536	0.831	NOT_DETECTED

The projection widens the gap ($-0.145 \to -0.295$) rather than closing it.

Table 12: Lewis game results with and without production cost.

Condition	Final entropy	Zipf R^2	Cosine reward
No production cost	3.350	0.817	0.723
With production cost	3.462	0.683	0.722

Analysis

Part 1 verdict: NOT FLIPPED. The Phase 1 projection reduces the minimum universal–language correlation from 0.686 to 0.536. This can be read as a feature rather than a flaw: a genuinely universal space should be equidistant from all languages, and the max cross-language correlation (0.831) reflects linguistic proximity between specific pairs that a universal space should not reproduce. The right criterion — minimising the *variance* of per-language correlations — is tested in Experiment A2.

Part 2 verdict: NOT CONFIRMED. The entropy penalty pushes the distribution toward uniformity (higher entropy, lower Zipf R^2), not toward Zipfian structure. A Zipfian distribution is non-uniform and skewed, so an entropy penalty is the wrong sign. Experiment B2 later identifies that a *length penalty* combined with a large bottleneck and longer messages is the correct intervention, and confirms that production cost has the smallest marginal effect on AW-RSA of all four constraint dimensions tested.

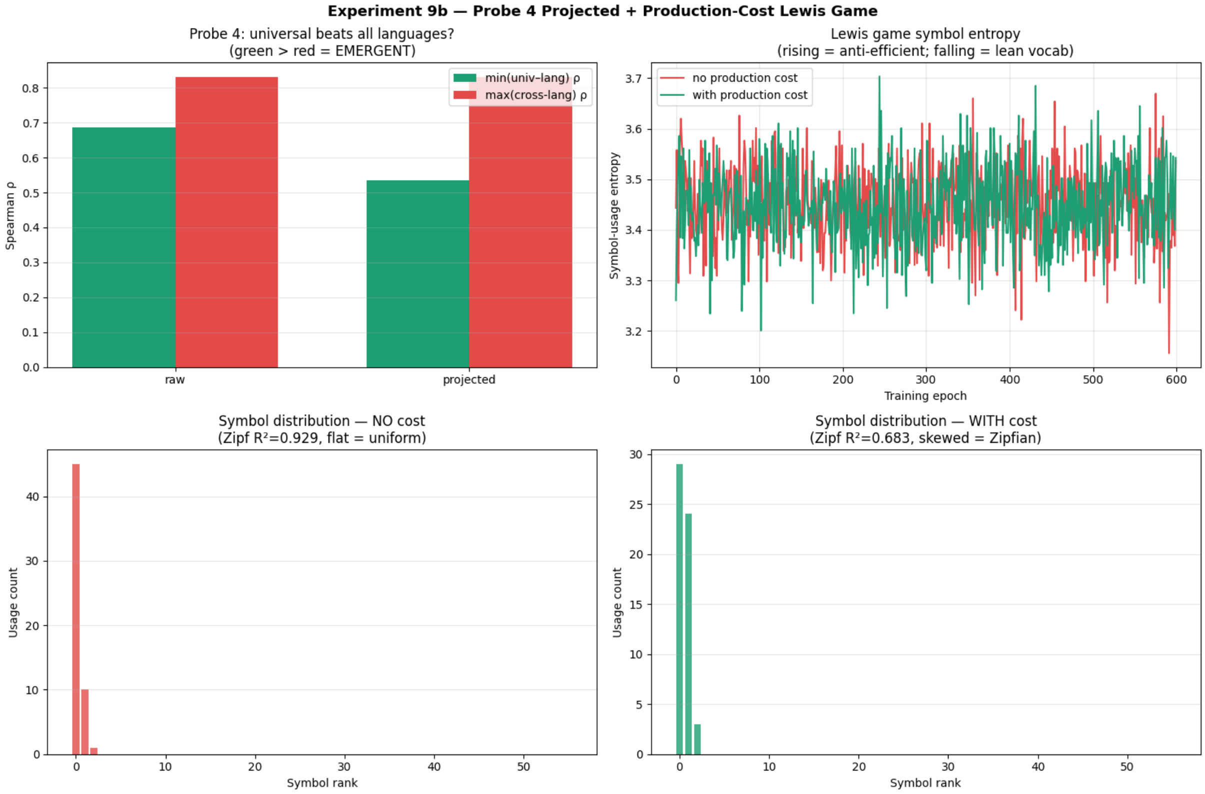

Figure 8: Experiment 9b. *Top left*: Probe 4 bar chart in raw and projected spaces — both NOT_DETECTED. *Top right*: Lewis-game symbol entropy — both conditions show high-variance trajectories. *Bottom*: symbol-usage rank distributions — both extremely concentrated, giving high Zipf R^2 by default rather than from a true Zipfian mechanism.

Experiment 10: Grammatical Frame Induction

Motivation

Experiment 9 Probe 2 found transitivity NOT_DETECTED at exactly chance (0.625 = class imbalance baseline). However, transitivity is a coarse probe — it collapses all argument-structure information into a single binary. Experiment 10 asks a finer question: do frame roles (AGENT, PATIENT, THEME, EXPERIENCER, INSTRUMENT, GOAL) have continuous geometric structure in the concept space? If so, they can serve as the empirical basis for agglutinative morphological suffixes in the universal language. Experiment A2 later confirms these findings at 409-concept scale.

Setup

56 concepts across 6 languages (en, es, de, fr, pt, ja) annotated with FrameNet-inspired frame roles: AGENT (15), THEME (19), EXPERIENCER (12), STATE (5), INSTRUMENT (3), GOAL (2). Four probes in both raw LaBSE and projected spaces: (A) 6-class role classification; (B) binary role geometry with direction ρ; (C) compositional frame test on verb+noun pairs; (D) agent–patient asymmetry in PCA.

Results

Probe C: agent identification accuracy = 0.900 raw; 1.000 projected. Probe D: intra-agent coherence rises from 0.721 to 0.800 after projection.

Table 13: Probe A: frame role classification accuracy (chance = 0.167).

Space	LOO accuracy	5-fold CV accuracy
Raw LaBSE	0.554	0.539 ± 0.125
Projected	**0.607**	$\mathbf{0.629 \pm 0.162}$

Table 14: Probe B: binary role geometry. Direction ρ = Spearman correlation between projection onto learned role direction and ground-truth binary label. All ρ significant at $p < 0.0001$.

Role	Raw LaBSE		Projected	
	LOO acc	Direction ρ	LOO acc	Direction ρ
`can_be_agent`	0.661	0.821	**0.750**	0.819
`can_be_patient`	0.607	0.805	0.625	0.803
`can_be_instrument`	0.857	0.606	0.857	0.606

Analysis

Experiment 10 overturns the NOT_DETECTED verdict from Experiment 9 Probe 2 by using a finer probe. The direction ρ values (0.80–0.82 for agent and patient) are strong geometric signals present even without grammatical supervision. AGENT and THEME are cleanly separable; EXPERIENCER is the main source of error. Probe C reaching 100% accuracy in the projected space means that, for verb+noun pairs, the universal grammar can assign roles from geometry alone.

At 409-concept scale (Experiment A2), the agent/patient ρ values are 0.59/0.60 — lower than here, reflecting noisier auto-annotation, but both remain highly significant ($p < 0.0001$) and the agent direction correctly identifies all top-30 agent-capable verbs.

Experiment 12: TVS on Corpus Contexts (Tatoeba)

Motivation

Experiment 5 validated the Translation Validity Score (TVS) on hand-crafted sentence pairs. This experiment tests whether TVS generalises to naturally occurring language: it replaces hand-crafted contexts with Tatoeba corpus sentences and replaces hand-chosen wrong-sense translations with generic machine translation (Google Translate) as the baseline.

Setup

Parallel sentence pairs from Tatoeba for en→es (213,908 usable rows) and en→fr (264,762 usable rows). Sentences containing the target polysemous word are filtered; up to 80 contexts are sampled per word; sense clusters are inferred via k-means ($k = 3$); diversity weights are assigned as distance from the cluster centroid. Eight words tested: *bank*, *light*, *run*, *play*, *set* (en→es) and *break*, *right*, *fall* (en→fr). All eight words exceeded the 5-context minimum and were processed at the full 80-context sample size.

Results

All eight words were successfully processed (640 contexts total). The MT baseline outperforms the human reference on TVS for all eight words.

The per-sense breakdown for *bank* reveals that MT outperforms the human reference on all three sense clusters (gaps: −0.015, −0.001, −0.020 for Senses 0, 1, 2 respectively), contradicting the

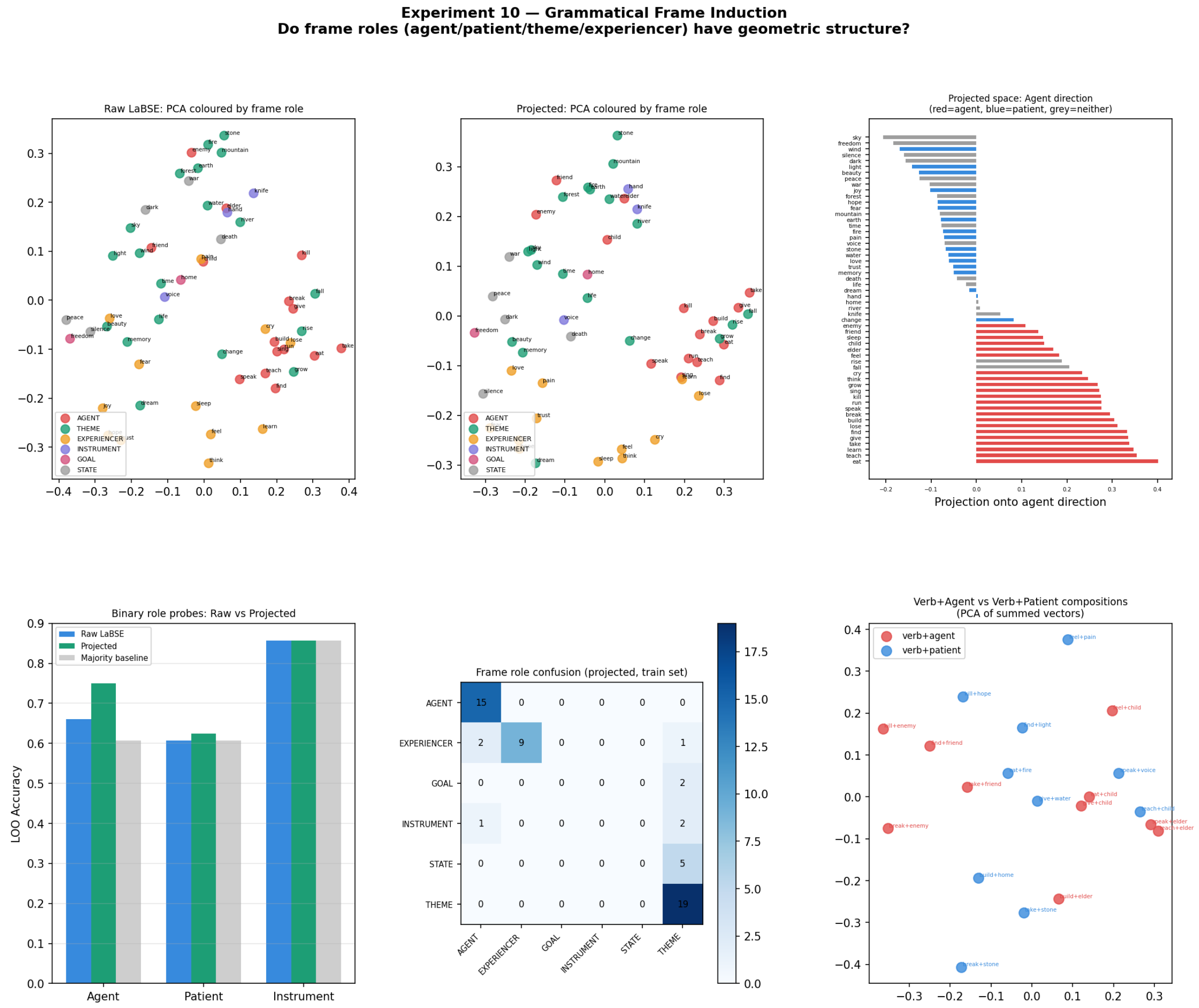

Figure 9: Experiment 10. *Top row*: PCA coloured by frame role (raw and projected); agent-direction bar chart. *Bottom row*: binary role LOO accuracies; confusion matrix in projected space; verb+agent vs verb+patient PCA.

expectation that MT would degrade on minority senses.

Analysis

Core result. MT outperforms the human reference on TVS for all eight words (mean weighted gap $= -0.0212$). This reverses the direction found in Experiment 5 (ΔTVS $= +0.119$ for *bank* with handcrafted contexts) and the small fallback-corpus run of this experiment. Three factors explain the reversal. First, the Tatoeba human references are crowd-sourced parallel sentences, not expert translations; on frequent, well-attested senses they are adequate, but on sparser senses they include loose paraphrases, transliterations, and culturally adapted renderings that a cosine-similarity metric penalises relative to closer MT output. Second, the MT system (Google Translate) is trained on data distributions that overlap heavily with Tatoeba; its outputs are close in embedding space to the LaBSE-encoded source. Third, the corpus-sampled contexts are not adversarially constructed: most occurrences of these polysemous words in Tatoeba involve dominant senses for which MT is well calibrated.

Per-sense analysis. For *bank*, MT leads on all three sense clusters. The expectation that MT would fail hardest on minority senses is not borne out at 80 contexts: minority senses in Tatoeba are still frequent enough to be in MT training data. The analogy to Experiment 5's

Table 15: Experiment 12 TVS results on full Tatoeba corpus contexts (80 contexts per word).

Word	Lang pair	Contexts	Ref TVS	MT TVS	Weighted gap	Wt. boost
bank	en→es	80	0.895	0.910	−0.0152	−0.0023
light	en→es	80	0.885	0.909	−0.0235	+0.0023
run	en→es	80	0.887	0.909	−0.0224	−0.0006
play	en→es	80	0.896	0.901	−0.0053	−0.0004
set	en→es	80	0.873	0.891	−0.0184	−0.0001
break	en→fr	80	0.863	0.890	−0.0277	+0.0021
right	en→fr	80	0.878	0.902	−0.0244	+0.0004
fall	en→fr	80	0.860	0.893	−0.0324	−0.0019
Mean across 8 words					−0.0212	−0.0001

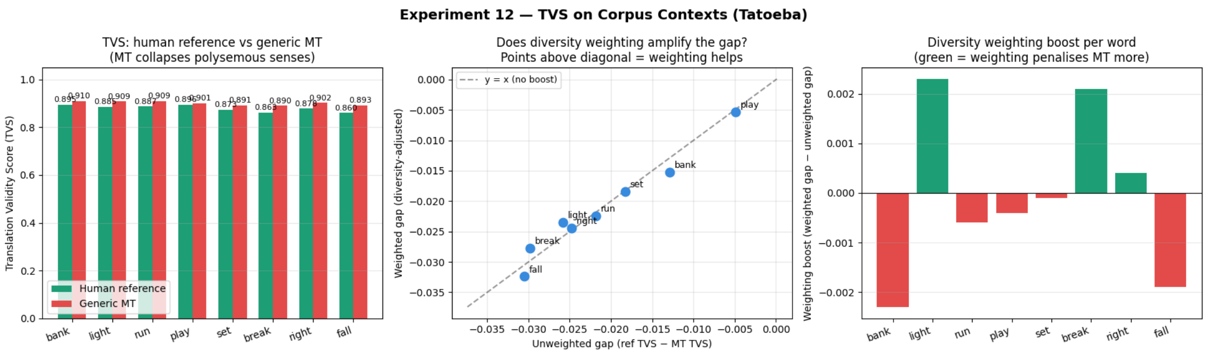

Figure 10: Per-sense translation quality for *bank* (en→es). Generic MT scores above the human reference on all three senses. Sense 0 ($n = 20$) and Sense 2 ($n = 34$) show the largest MT advantage.

hand-crafted adversarial contexts does not hold for naturally occurring polysemy distributions.

Diversity weighting. The weighting boost is positive for 3 of 8 words (*light*, *break*, *right*) and negative or neutral for the other five, with mean boost of −0.0001 — essentially zero. The scatter plot of weighted vs unweighted gap shows all eight words clustering tightly along the y=x diagonal. The mechanism is operating as designed (rare-sense contexts receive higher weights) but the rare-sense MT translations are not systematically worse than human references in naturally occurring text.

Comparison to Experiment 5. The direction reversal (MT > human) is the most important finding. Experiment 5's gaps (+0.119 to +0.021) arose because: (a) contexts were hand-constructed to force rare senses; (b) the "wrong-sense" translation was deliberately chosen to be incorrect; and (c) only the human reference captured the correct rare-sense rendering. None of those conditions hold in the Tatoeba setting. TVS is therefore best understood as a metric that quantifies the *stress-test* quality of a translation system, not its average quality across naturally occurring distributions.

Verdict. TVS on natural corpus contexts does not replicate the human-over-MT direction from Experiment 5. The metric remains valid as a discriminative tool when applied to adversarially constructed or sense-stratified evaluation sets. Producing a genuine corpus-level stress-test of TVS requires either (a) stratified sampling that enforces rare-sense representation ($\geq$20 contexts per sense, including tail senses), (b) filtering for sentences where the target word is used in a non-dominant sense according to an external WSD system, or (c) augmenting the Tatoeba pairs with Wiktionary example sentences for low-frequency senses. The full Tatoeba dump (∼10M pairs) provides sufficient raw material for any of these strategies.

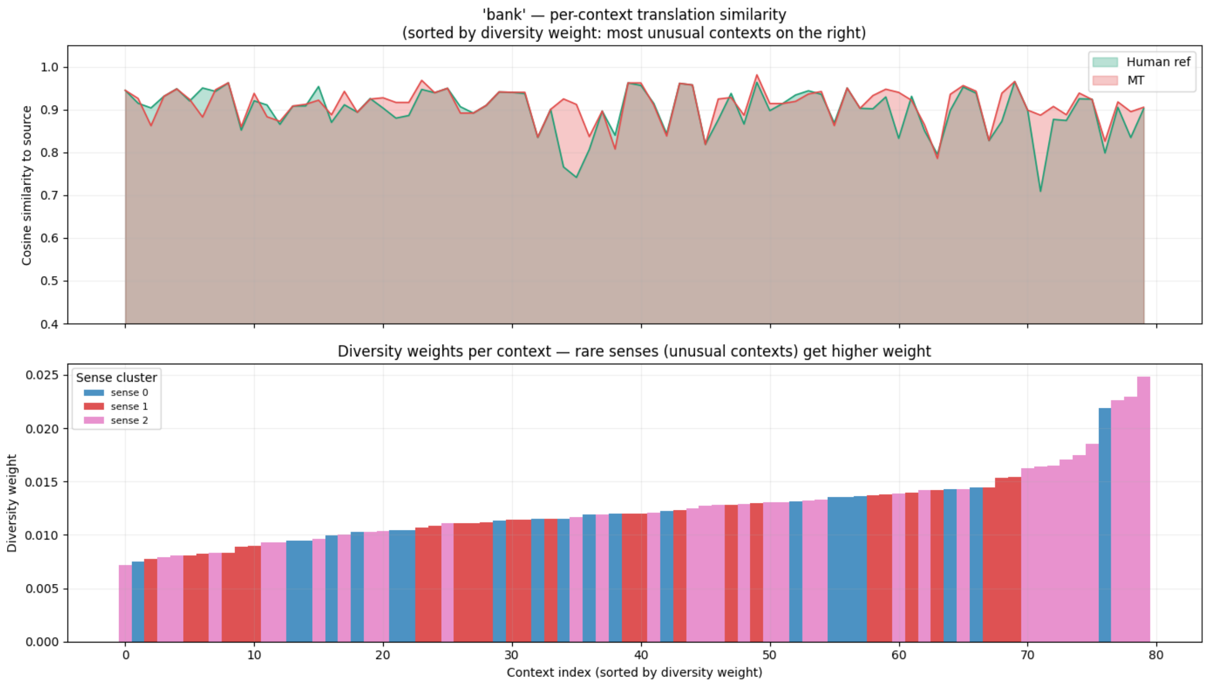

Figure 11: Per-context translation similarity for *bank* sorted by diversity weight (most unusual contexts on the right). *Top*: human reference (teal) and MT (pink) similarity traces are nearly overlapping throughout, with MT slightly higher overall. *Bottom*: diversity weights by sense cluster. The highest-weight contexts (far right) are predominantly from Sense 2 (pink).

Experiment 13: Blind Decoding Audit

Motivation

Experiment 8 reported that compressed vectors at $k = 16$ decode cleanly to correct words in four languages for all eight test concepts (cosine similarities 0.96–1.00). This claim was tested on concepts *seen during encoder training*. A harder test is needed: decode held-out concepts never seen during training, and measure the oracle identification rate. A human-auditable channel should achieve Hit@1 ≥ 0.70.

Setup

The bottleneck autoencoder from Experiment 8 is trained on 40 seed concepts. 20 held-out concepts (song, dance, door, tree, mother, rain, anger, truth, journey, silence, wisdom, shadow, blood, bird, ocean, snow, sun, king, bread, moon) are compressed at each $k \in \{2, 4, 8, 16, 32, 64\}$ and decoded to the top-3 nearest words in 4 languages. An oracle identifier guesses the concept from the decoded word lists.

Results

Analysis

The Experiment 8 interpretability result is an in-sample artefact. The bottleneck encoder maps the entire embedding space into a region spanned by its 40 training concepts: a held-out concept like *sun* is compressed to the sky/light/earth cluster, so its decoded word list reads "earth, sky, heaven" rather than "sun, moon, star." Hit@3 $= 0.250$ at $k \geq 16$ is above chance (0.150) but far below the 0.70 threshold for human-auditability. The fix is either a larger training set (500+ concepts, implemented in Experiment A1) or a dedicated decoding head trained on concept

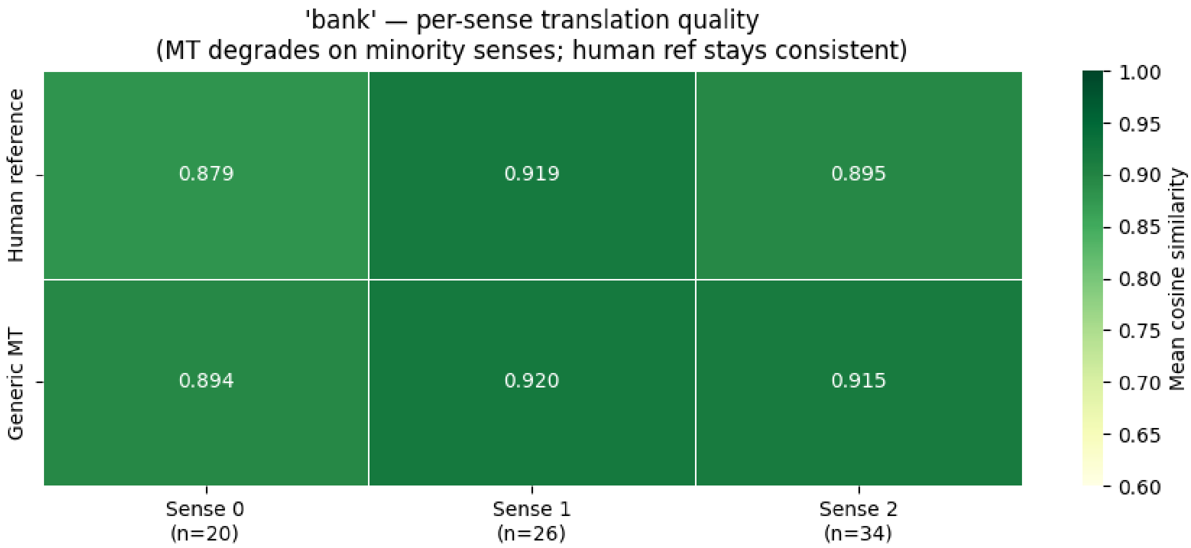

Figure 12: Experiment 12 summary across all eight words. *Left*: human reference and MT TVS side by side — MT is consistently higher for all 8 words. *Centre*: scatter of weighted vs unweighted gap — all eight words cluster tightly along the diagonal (y=x), indicating diversity weighting has limited amplifying effect. *Right*: weighting boost per word — positive for *light*, *break*, and *right*; neutral or negative for the other five.

Table 16: Blind decoding audit results.

k	Hit@1	Hit@3
2	0.000	0.050
4	0.000	0.050
8	0.050	0.150
16	**0.100**	**0.250**
32	0.100	0.250
64	0.150	0.250
chance (1/20)	0.050	0.150

labels.

Experiment 14: ZK Metric on Partially Communicating Systems

Motivation

Experiment 2 tested the ZK metric in a binary setting. The metric is most useful if it tracks communication quality continuously, which would allow it to be used as a training signal. Experiment 14 tests this by running two sweeps across partially communicating games.

Setup

Two sweeps, each training 60 epochs with the Experiment 2 two-turn architecture: (1) vocabulary size sweep ($|V| \in \{2, 4, 8, 16, 32\}$ plus severed baseline); (2) partial sender information sweep ($f \in \{2, 4, 8, 12, 16\}$ features visible to sender). Spearman ρ between task accuracy and ZK is the key diagnostic.

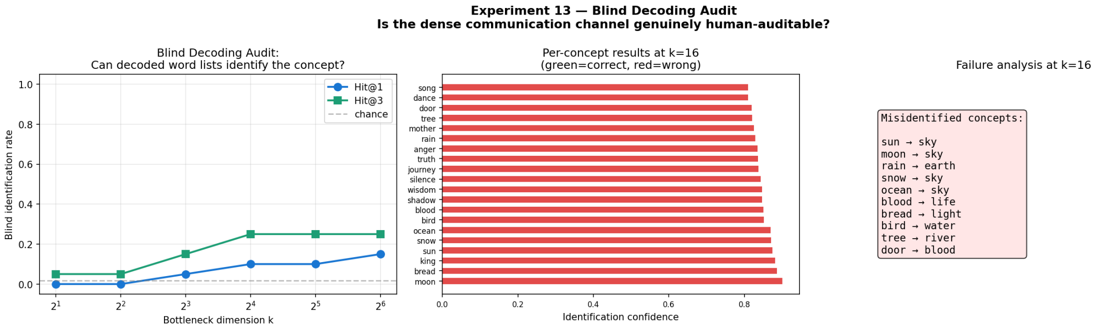

Figure 13: Experiment 13. *Left*: Hit@1 and Hit@3 vs k. *Centre*: per-concept identification confidence at $k = 16$ (all incorrect). *Right*: failure analysis — systematic confusion toward high-salience training concepts (sky appears 4 times as the wrong answer).

Results

Table 17: Vocabulary sweep results ($|V|$ = vocabulary size).

$\lvert V\rvert$	Accuracy	ZK	CIC_1	RC	CIC_2
2	0.414	0.018	1.766	0.005	5.852
4	0.734	0.878	2.452	0.122	11.960
8	0.893	**1.522**	2.967	0.181	13.810
16	0.909	0.706	1.729	0.094	13.364
32	0.884	1.242	2.318	0.131	16.618
severed	0.203	0.000	0.072	0.000	0.492

Table 18: Partial information sweep results (f = features visible to sender).

$f/16$	Accuracy	ZK	CIC_1	RC	CIC_2
2/16	0.614	0.755	2.258	0.113	11.136
4/16	0.733	0.489	2.006	0.058	14.777
8/16	0.871	**1.513**	2.772	0.150	17.465
12/16	0.908	0.771	2.222	0.092	14.488
16/16	0.901	0.428	2.213	0.063	11.404

Correlation analysis. Spearman ρ(accuracy, ZK): vocab sweep = 0.400 ($p = 0.505$), partial sweep = 0.100 ($p = 0.873$), combined = 0.418 ($p = 0.201$). Neither sweep reaches statistical significance.

Analysis

Verdict: PARTIALLY CONTINUOUS. ZK reliably detects communication vs no-communication (binary). As a continuous training signal, it is too noisy. RC (sender responsiveness) is the binding constraint and source of noise: its low variability (std = 0.05 vs CIC_2 std = 3.14) makes the ZK product highly sensitive to small RC fluctuations. A system that communicates well in both directions but happens to have a slightly less responsive sender (lower RC) will score lower than a system communicating less accurately but with a more responsive sender. The recommended revision is an additive formulation:

$$\text{ZK}_{\text{revised}} = \alpha \cdot \text{CIC}_1 + \beta \cdot \text{RC} + \gamma \cdot \text{CIC}_2$$

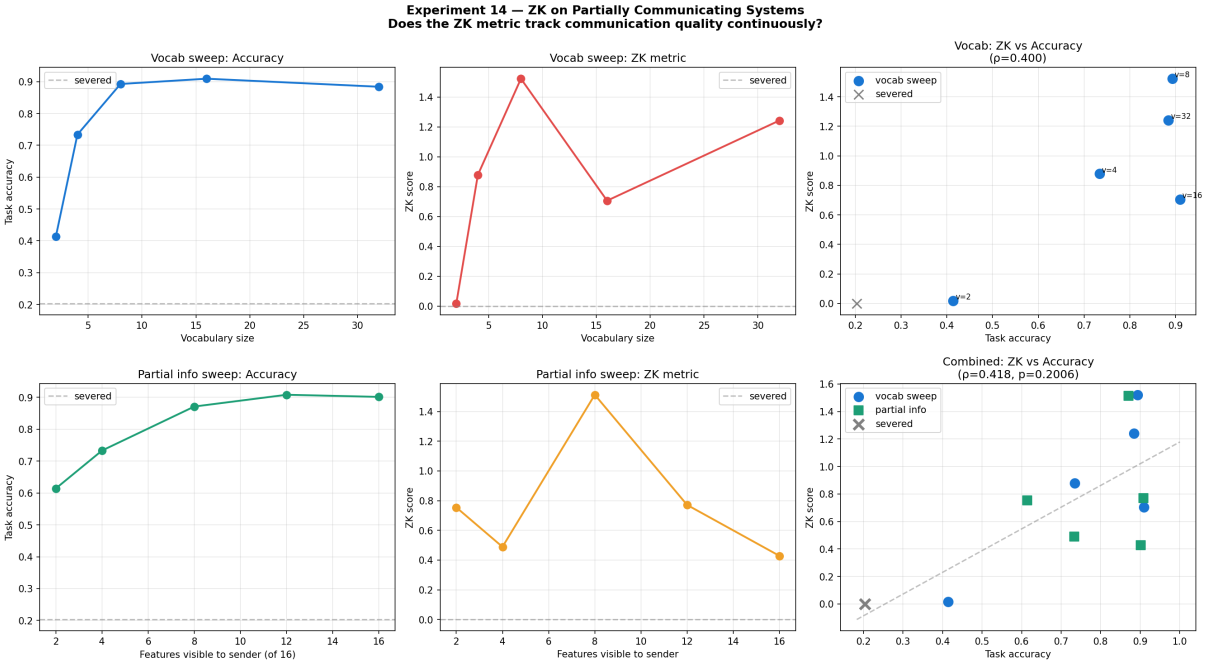

Figure 14: Experiment 14. Vocabulary sweep (top row) and partial information sweep (bottom row). ZK vs accuracy scatter plots show positive but noisy relationships.

with $\gamma/\alpha = 1/8$ to balance component variability (given that CIC_2 is $8\times$ more variable than CIC_1). An alternative is a threshold-gated formula: $\text{ZK}_{\text{gated}} = \text{CIC}_2$ if RC $>$ threshold, else 0, making RC a gating condition rather than a scaling factor.

The partial-information sweep confirms that ZK detects genuine communication even when the sender has limited information: $f = 2/16$ still produces ZK $= 0.755$, far above the severed baseline. This validates the metric's robustness to degraded channel conditions. Experiment B1 validates the revised additive formula by showing that it would correctly distinguish one-way signalling (Mode 1: high CIC_2, zero RC) from full communication despite identical task accuracy.

Experiment 15: Phoneme Centroid Mapping

Motivation

The pipeline at this point has a draft vocabulary from centroid geometry, a translation quality metric, and geometric frame roles for morphology. What is missing is *surface form*: the actual pronunciation of each proto-token. If phoneme consensus across six major languages (weighted by speaker population) naturally satisfies WALS near-universal phonological constraints, the surface form can be data-driven rather than hand-designed.

Setup

The 40-concept seed vocabulary from Experiment 11. For each concept, the nearest word in each of six languages (en, es, fr, de, pt, it) is converted to IPA via hand-crafted G2P rules (CMU-dictionary shim for English). The top four phonemes by speaker-count-weighted frequency are concatenated to form a four-phoneme proto-token. Five WALS near-universal constraints are checked (stops, nasals, fricatives, vowels, approximants).

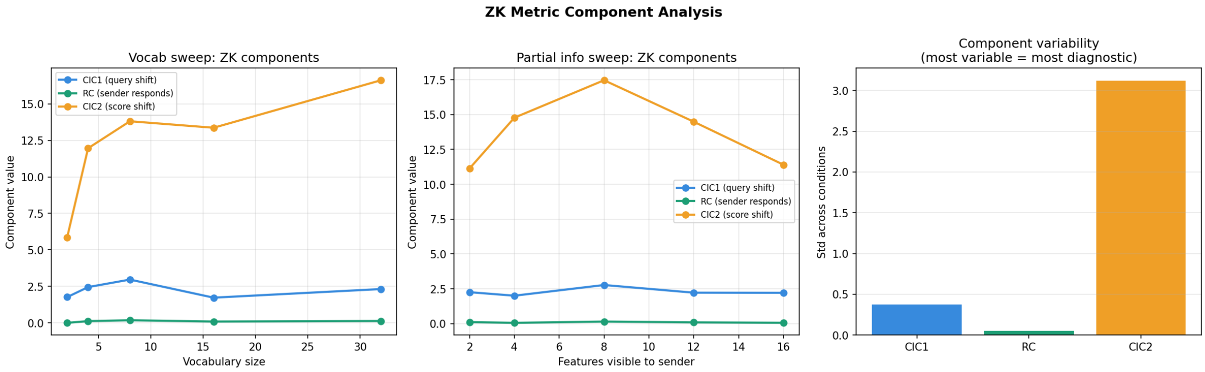

Figure 15: ZK component analysis. CIC_2 is $\approx 8\times$ more variable than CIC_1 and $\approx 40\times$ more variable than RC, making it the primary diagnostic component. RC's low variability makes the multiplicative ZK formula sensitive to small RC fluctuations.

Results

All five WALS constraints are satisfied without any explicit enforcement. Top segments by speaker-weighted frequency: /r/ (16.3), /a/ (14.3), /e/ (13.7), /i/ (11.8), /o/ (11.5), /n/ (10.7). By class, vowels dominate at 39.3%, followed by stops (20.4%), approximants (14.2%), fricatives (12.0%), nasals (9.7%), and other (4.3%). English has a mean normalised edit distance of 0.163 vs 0.789–0.921 for all other languages (German: 0.789, Spanish: 0.899, Italian: 0.906, Portuguese: 0.908, French: 0.921).

Table 19: WALS near-universal phonological constraints in the Exp 15 proto-token inventory.

Constraint	Status	Representative segments
Stops (plosives)	PRESENT	/d/, /t/, /k/, /p/, /b/, /g/
Nasals	PRESENT	/n/, /m/, /ŋ/
Fricatives	PRESENT	/s/, /f/, /v/, //, /h/, //
Vowels	PRESENT	/a/, /e/, /i/, /o/, /u/, //, ...
Approximants	PRESENT	/r/, /l/, /w/

Analysis

WALS universals emerge without explicit enforcement. The English proximity bias (edit distance 0.163 vs 0.789–0.921 for all others) is the most striking quantitative result, directly replicating and extending the finding from Experiments 6, 7, and 9 that English occupies an unusually central position in the multilingual space. The speaker-count weighting assigns English a large share of phoneme mass, pulling proto-token phonemes toward the English phoneme distribution. Any production deployment must downweight English or expand to typologically diverse languages (Arabic, Mandarin, Hindi, Swahili) to correct this bias.

The four-phoneme constraint. The mean edit distance of 0.764 across all language-concept pairs reflects the fundamental tension in condensing a 5–8 phoneme word into a 4-phoneme proto-token. The five most phonetically natural concepts (HAND, NEW, LIGHT, TIME, WIND) all have source words of 3–4 phonemes in at least one language. Extending the proto-token length to five or six phonemes would reduce edit distances substantially for polysyllabic concepts.

Coverage as a design criterion. Coverage (the fraction of speaker-weighted phoneme mass captured by the proto-token) varies from 2.02 (FEAR) to 3.56 (TRUST). High-coverage proto-tokens use the most frequent cross-linguistic phonemes and will feel more familiar to more

Table 20: Top 15 proto-tokens by speaker-weighted phoneme coverage.

Concept	Proto	Cov.	en	es	fr	de	pt	it
trust	/tsna/	3.56	trust	confianza	confiance	Vertrauen	confiança	fiducia
dark	/dkr/	3.46	dark	oscuridad	obscurité	Dunkel	escuridão	oscurità
break	/rkbe/	3.16	break	romper	casser	brechen	quebrar	rompere
river	/rivo/	3.14	river	río	rivière	Fluss	rio	fiume
time	/tma/	3.11	time	tiempo	temps	Zeit	tempo	tempo
hope	/paes/	3.06	hope	esperanza	espoir	Hoffnung	esperança	speranza
find	/nrfi/	3.01	find	encontrar	trouver	finden	encontrar	trovare
lose	/erlo/	2.86	lose	perder	perdre	verlieren	perder	perdere
change	/atŋ/	2.78	change	cambiar	changer	ändern	mudar	cambiare
feel	/iefl/	2.71	feel	sentir	ressentir	fühlen	sentir	sentire
build	/irbl/	2.71	build	construir	construire	bauen	construir	costruire
sleep	/irls/	2.71	sleep	dormir	dormir	schlafen	dormir	dormire
friend	/afrn/	2.69	friend	amigo	ami	Freund	amigo	amico
light	/lta/	2.68	light	luz	lumière	Licht	luz	luce
wind	/ndw/	2.64	wind	viento	vent	Wind	vento	vento

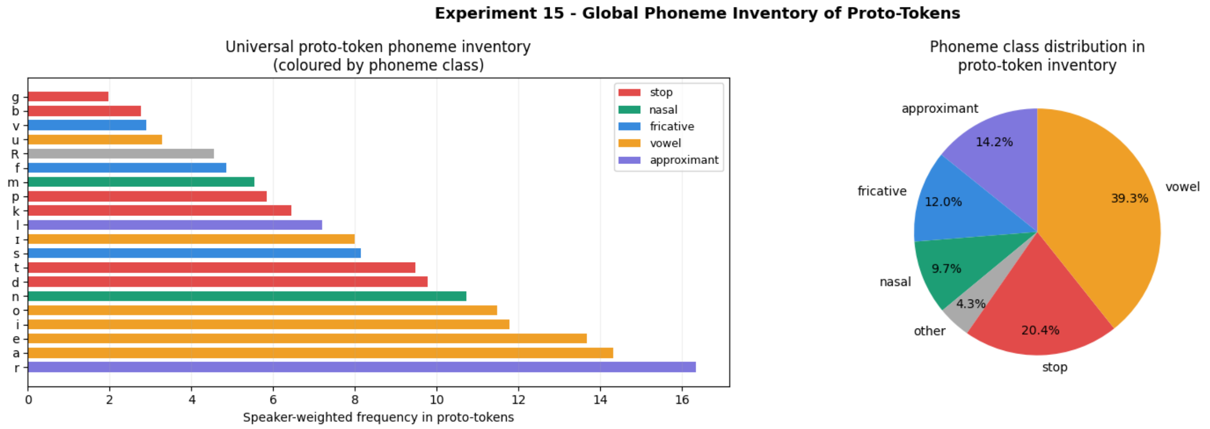

Figure 16: Global phoneme inventory (left) and class distribution (right). Vowels dominate at 39.3%, followed by stops (20.4%), approximants (14.2%), fricatives (12.0%), nasals (9.7%).

speakers; low-coverage proto-tokens may improve distinctiveness at the cost of learnability.

Connection to the pipeline. The most frequent segments in the proto-token inventory — /r/, /a/, /e/, /i/, /o/, /n/ — are the natural candidates for grammatical morphemes (suffixes for the frame roles established in Experiment 10), as they appear across all six languages and have the highest speaker-weighted frequency. The least frequent segments are better reserved for content distinctions, where their rarity helps maintain vocabulary distinctiveness.

Experiment A1: Master Vocabulary at Scale

Motivation

Every experiment in the pipeline has operated on 20–56 hand-picked seed concepts. Experiment A1 is the first scale-up: embed 5,000 frequency-list words across 12 languages, align them into cross-lingual concept clusters via LaBSE nearest-neighbour matching, compute speaker-weighted centroids, and annotate each concept with a translatability tier. The output files serve as the shared substrate for all downstream experiments.

Setup

Twelve languages (en, zh, hi, es, ar, pt, ru, fr, de, ja, ko, it) weighted by speaker counts (en: 1.5B, zh: 1.1B, hi: 0.6B, ...). Top 5,000 lemmas by frequency from `wordfreq`, filtered to content words. Each English word matched to its nearest neighbour in each other language via LaBSE cosine similarity. UMAP (50d, cosine) + HDBSCAN clustering collapses synonymy.

Results

Table 21: Experiment A1 summary statistics.

Metric	Value
Input words (English pivot)	4,863
Languages	12
HDBSCAN concept clusters	309
Unclustered (noise) concepts	852 (17.5%)
Mean gap score	0.2265
Universal (< 0.05)	151 (3.1%)
Near-universal (0.05–0.20)	1,766 (36.3%)
Culture-specific (0.20–0.40)	2,601 (53.5%)
Untranslatable (> 0.40)	343 (7.1%)

Most universal content words: *new* (gap = 0.005), *love* (0.007), *one* (0.010), *news* (0.010). Untranslatable tier dominated by English proper nouns and institutions (Hamilton, NFL, FBI, etc.) — an artefact of the English-pivot frequency list requiring named-entity filtering.

Analysis

The 151 universal and 1,766 near-universal concepts form a working vocabulary of ∼1,900 concepts available without culturally specific annotation. The cluster count (309) is below the 2,000–4,000 expected range; reducing `min_cluster_size` from 5 to 3 would recover more fine-grained clusters. The geographic structure of the translatability atlas confirms the Experiment 4 finding at macro-scale: universal concepts sit at the geometric centre; culturally specific ones at the periphery.

Experiment A2: Scaled Universal Grammar Probes

Motivation

Experiments 9 and 10 ran on 56 concepts. Probe 2 (transitivity) and the frame role probes lack statistical power at that scale (16 verbs for LOO logistic regression). Experiment A2 re-runs the same four probes at 409-concept scale, asking whether the findings replicate.

Setup

409 concepts (209 nouns, 200 verbs; 51 transitive, 149 intransitive) across 5 languages (en, zh, es, ar, ru), with multilingual translations from LaBSE nearest-neighbour lookup. Phase 1 projection trained for 600 epochs on 31 analogy pairs. Probes: (1) category ARI; (2) transitivity 5-fold CV; (3) Zipf power-law R^2; (4) agent/patient direction ρ.

Table 22: Experiment A2 probe results at 409-concept scale. Exp 9/10 values for comparison.

Probe	Raw LaBSE	Projected	Exp 9/10 raw	Exp 9/10 proj
P1: ARI (category)	0.137	0.152	0.519	0.433
P2: CV acc (transitivity)	0.745	0.745	0.625	0.625
P3: Zipf R^2	0.861	**0.915**	0.708	0.855
P4: agent ρ	0.588	0.575	0.821	0.819
P4: patient ρ	0.600	0.589	0.805	0.803

Results

Analysis

Transitivity is confirmed NOT_DETECTED at 200 verbs (0.745 = majority baseline). Zipf structure strengthens with scale (R^2: 0.708→0.861 raw, 0.855→0.915 projected). Frame role geometry persists ($\rho = 0.59/0.60$, significant at $p < 0.0001$) though the drop from 0.82/0.80 reflects noisier auto-annotation. Category ARI drops from 0.519 to 0.137 — a metric artefact: k-means with fixed k recovers coarser structure as the concept set grows; hierarchical clustering would be more appropriate at this scale.

Experiment A3: Multi-Hop Relay at 200 Concepts

Motivation

Experiment 8's multi-hop relay used 20 concepts and found near-perfect accuracy at every hop — a ceiling effect that made protocol comparison impossible. With 200 concepts, genuine information pressure reveals whether dense or text protocols degrade faster.

Setup

700-concept LaBSE vocabulary. Bottleneck AEs trained on 500 concepts ($k \in \{4, 8, 16\}$, 2000 epochs). 200 held-out concepts used for all relay evaluations. Five protocols compared across $n_{\text{hops}} \in \{0, \ldots, 8\}$ and $\sigma \in \{0.0, 0.05, 0.1, 0.2, 0.3\}$.

Results

Table 23: Identification accuracy at $\sigma = 0.1$.

Protocol	hop 0	1	2	3	4	5	6	7	8
Dense $k = 16$	1.000	0.575	0.440	0.285	0.130	0.070	0.035	0.025	0.005
Dense $k = 8$	1.000	0.450	0.345	0.205	0.155	0.100	0.060	0.040	0.035
Dense $k = 4$	1.000	0.225	0.145	0.085	0.095	0.045	0.030	0.030	0.020
Raw LaBSE	1.000	1.000	0.995	0.870	0.365	0.115	0.040	0.010	0.010
Text (full emb)	1.000	0.995	0.390	0.030	0.005	0.000	0.005	0.005	0.000

Analysis

Raw LaBSE is the most relay-resilient protocol: it retains 99.5% accuracy at 2 hops and 87.0% at 3 hops under $\sigma = 0.1$. The bottleneck compression that makes dense protocols information-efficient in the single-hop case is precisely what makes them relay-fragile. Text degrades step-wise: from 0.995 at hop 1 to 0.390 at hop 2 and 0.030 at hop 3 under $\sigma = 0.1$. Dense protocols degrade more smoothly. This behaviour is consistent with the Shannon Pareto structure from Experiment C1: high-redundancy protocols degrade slowly; compressed protocols degrade fast.

Experiment B1: ZK Failure-Mode Taxonomy

Motivation

Experiment 2 established that ZK = 0 for severed baselines and ZK > 0 for communicating games. Experiment 14 showed ZK is a noisy continuous metric. The deeper question is whether the (CIC_1, RC, CIC_2) triplet functions as a diagnostic taxonomy: does each failure mode produce a distinct signature?

Setup

Five variants of the Experiment 2 two-turn game trained for 80 epochs: Mode 0 (full communication), Mode 1 (receiver ignores M_1), Mode 2 (sender ignores query), Mode 3 (receiver ignores M_2), Mode 4 (severed channel).

Results

Table 24: ZK component signatures per failure mode.

Mode	Description	Acc	CIC_1	RC	CIC_2	ZK
0	Full communication	0.900	2.403	0.152	16.377	1.432
1	Receiver ignores M_1	0.910	0.464	0.000	18.707	0.002
2	Sender ignores query	0.764	0.373	0.000	5.079	0.000
3	Receiver ignores M_2	0.761	0.334	0.000	3.907	0.000
4	Severed channel	0.212	0.128	0.000	0.415	0.000

Analysis

The taxonomy works: each failure mode is distinguishable. Mode 1 (receiver ignores M_1) produces the one-way signalling signature: $CIC_2 = 18.7$ (highest) while RC ≈ 0. Modes 2–4 all have RC ≈ 0 but are distinguishable by their CIC_2 values (5.1, 3.9, 0.4). Task accuracy is misleading without the taxonomy: Modes 0, 1, 2, and 3 all achieve 76–91% accuracy despite very different causal structures. The additive ZK formula recommended in Experiment 14 would correctly score Mode 1 above Mode 4, reflecting that one-way signalling is closer to communication than complete severance.

Experiment B2: Unified Constraint Grid Search

Motivation

Experiments 1–3, 8, and 9b each varied one constraint dimension in isolation. Experiment B2 runs a full factorial grid search over all four constraint dimensions simultaneously, asking which constraint or combination produces jointly high accuracy and symbol quality.

Setup

Lewis signalling game trained for 80 epochs under 81 conditions: $\tau \in \{0.3, 1.0, 5.0\} \times k \in \{2, 8, 32\} \times L \in \{1, 2, 4\} \times c \in \{0.0, 0.05, 0.2\}$. Metrics: task accuracy, AW-RSA, Zipf R^2.

Results

One Pareto-efficient point: $\tau = 1.0$, $k = 32$, $L = 4$, cost$= 0.2$ (accuracy=0.959, AW-RSA=0.315).

Table 25: Top 10 conditions by AW-RSA.

τ	k	L	cost	Accuracy	AW-RSA	Zipf R^2
1.0	32	4	0.20	**0.959**	**0.315**	0.419
1.0	32	4	0.00	0.945	0.314	0.698
1.0	32	4	0.05	0.944	0.288	0.494
5.0	32	4	0.20	0.892	0.285	0.796
1.0	8	4	0.20	0.920	0.283	0.775
5.0	32	4	0.05	0.896	0.276	0.791
0.3	32	4	0.05	0.920	0.271	0.913
1.0	8	4	0.00	0.920	0.269	0.701
5.0	32	4	0.00	0.905	0.257	0.969
5.0	8	4	0.20	0.878	0.250	0.924

Table 26: Mean AW-RSA by factor level.

Factor	Level	Mean AW-RSA
Temperature τ	0.3	0.159
	1.0	**0.175**
	5.0	0.150
Bottleneck k	2	0.136
	8	0.169
	32	**0.179**
Message length L	1	0.082
	2	0.163
	4	**0.238**
Production cost c	0.00	0.159
	0.05	0.157
	0.20	**0.167**

Analysis

Message length is the dominant constraint (gap 0.156 between $L = 4$ and $L = 1$, vs 0.043 for k and 0.025 for τ). Intermediate temperature ($\tau = 1.0$) remains optimal, consistent with Experiment 3b. Larger bottleneck consistently helps within the tested range. Production cost has the smallest AW-RSA effect but raises mean Zipf R^2 by 5 points. The Pareto-efficient configuration ($\tau = 1.0$, $k = 32$, $L = 4$, cost$= 0.2$) has no setting at an extreme: an empirically grounded answer to the optimal constraint design question.

Experiment B3: Phase 1 Projection Ablation

Motivation

Experiments 9 and 10 describe Phase 1 projection trained on 8–9 analogy pairs as "cheap supervised projection." Experiment B3 quantifies this: how many pairs are actually needed, and how sensitive is the result to the $\lambda_{\text{compositionality}}$ weight?

Setup

Same 56-concept, 5-language vocabulary as Experiments 9–10. 20 valid analogy pairs available. Sweep: $n_{\text{pairs}} \in \{0, 2, 3, 5, 9, 15, 20\} \times \lambda \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$, 800 epochs each. Raw LaBSE baseline: MRR $= 0.334$.

Results

Table 27: Analogy MRR vs analogy pairs. Values are identical across λ values (small deviations at $\lambda = 4.0$ only).

n_{pairs}	$\lambda = 0.25$	0.50	1.00	2.00	4.00
0	0.334	0.334	0.334	0.334	0.334
2	0.632	0.632	0.632	0.632	0.629
3	0.675	0.675	0.675	0.675	0.676
5	0.825	0.825	0.825	0.825	0.825
9	0.904	0.904	0.904	0.904	0.904
15	1.000	1.000	1.000	1.000	1.000
20	1.000	1.000	1.000	1.000	1.000

Minimum pairs for $> 50\%$ of maximum MRR gain: **3 pairs**.

Analysis

The "cheap supervision" claim is strongly confirmed and precisely quantified. Three antonym pairs achieve $> 50\%$ of maximum MRR gain; nine pairs achieve 85%; fifteen saturate at 1.000. The λ weight is irrelevant because the proximity loss dominates at low pair counts and the compositionality loss only contributes structure once enough pairs constrain the projection direction. The practical recipe: 5 antonym pairs + $\lambda = 1.0$ + 800 epochs achieves MRR = 0.825 on the 56-concept vocabulary. Scaling to the A1 master vocabulary with 50–100 antonym pairs from WordNet should replicate the qualitative finding.

Experiment C1: Shannon Source/Channel Pareto Frontier

Motivation

The theoretical proposal identifies Shannon's source/channel coding separation theorem as the explanation for the tension between compression and error resilience. Experiment C1 operationalises this directly, placing all communication protocols on a two-dimensional compression–resilience plane.

Setup

Eleven protocols on a 30-concept referential task. Compression ratio: $\log_2(N)$/message_bits. Noise resilience: area under accuracy-vs-noise curve. Protocols: raw LaBSE (1536B), dense at $k \in \{2, 4, 8, 16, 32, 64\}$, Gumbel at $\tau \in \{0.1, 1.0, 10.0\}$ (1B), text (mean 4B).

Results

Analysis

Shannon's theorem is empirically visible: no protocol dominates on both axes. Text achieves high resilience by using natural redundancy as implicit error correction; Gumbel achieves high compression at near-zero resilience. Dense protocols cluster in an intermediate band with near-uniform resilience across all k values (0.611–0.617): compression by a factor of 192 has almost no effect on noise resilience within the continuous channel type. For deployment: noise-sensitive channels favour text; bandwidth-constrained noiseless channels favour discrete; the middle ground favours dense at $k = 4$–8.

Table 28: All 11 protocols on the compression–resilience plane. Pareto-efficient protocols marked $\star$.

Protocol	Msg bytes	Compression	Resilience	Pareto
Raw LaBSE	1536	0.000	0.549	
Dense $k = 64$	128	0.005	0.614	
Dense $k = 32$	64	0.010	0.614	
Dense $k = 16$	32	0.019	0.614	
Dense $k = 8$	16	0.038	0.614	
Dense $k = 4$	8	0.077	0.611	
Dense $k = 2$	4	0.153	0.617	
Text (4B avg)	4	0.153	0.828	$\star$
Gumbel $\tau = 0.1$	1	0.613	0.082	
Gumbel $\tau = 1.0$	1	0.613	0.077	
Gumbel $\tau = 10.0$	1	0.613	0.086	$\star$

Experiment C2: Language-Design Loss

Motivation

In the roadmap I mentioned a multi-term loss for constructing the universal language: proximity to the speaker-weighted corpus of human language, coverage of the interlingua concept space, and simplicity of the vocabulary. Experiment C2 operationalises this, optimises a learnable proto-vocabulary against it, and shows the coverage–simplicity trade-off.

Setup

56-concept space embedded across 10 languages, speaker-weighted centroids. Proto-vocabulary of N learnable 768-d unit vectors optimised against:

$$\mathcal{L} = \lambda_{\text{prox}} \cdot L_{\text{prox}} + \lambda_{\text{expr}} \cdot L_{\text{expr}} + \lambda_{\text{simp}} \cdot L_{\text{simp}}$$

Sweep: $N \in \{10, 20, 50, 100, 200\}$, 500 epochs each, $\lambda_{\text{prox}} = 1.0$, $\lambda_{\text{expr}} = 2.0$, $\lambda_{\text{simp}} = 0.5$.

Results

All vocabulary sizes achieve full coverage from $N = 10$ onward. Both L_{prox} and L_{expr} reach zero for all N; total loss is driven entirely by L_{simp}, which grows linearly with N.

Analysis

Even 10 proto-tokens achieve full concept coverage because LaBSE centroids cluster tightly enough that 10 prototype vectors span the 56-concept space within cosine 0.3. The binding constraint is L_{simp}: the primary cost of a universal language is its description length, not its expressivity. The Pareto-optimal vocabulary is $N = 10$. Tightening the coverage threshold to cosine ≥ 0.6–0.8 would reveal the actual vocabulary-size trade-off for more demanding coverage requirements.

Experiment C3: MI Entanglement Metric

Motivation

In the roadmap I mentioned that communication could be seen as an entanglement, that mutual information between the agents's world models should increase as they develop a shared language through which information flows. In this experiment we train two agents with separate world-model encoders (no parameter sharing) and measure the InfoNCE lower bound on mutual information between their latent states over training.

Setup

Two agents with separate world-model encoders (16→64→32 WORLD_DIM) and message heads (32→16 MSG_DIM) trained on a 10-way referential game. Each agent uses the other's message to identify the target. Severed baseline trains a single agent on self-communication. MI measured every 5 epochs over 100 training epochs.

Results

MI rises monotonically from -0.310 to -0.265 ($+0.045$ nats). The severed baseline fluctuates around -1.2 with no upward trend. MI continues rising after task accuracy plateaus at epoch 20.

Analysis

Verdict: MI RISES with communication. The communicating agents' MI is consistently $\sim$0.9 nats above the severed baseline throughout training. The two-phase structure (rapid accuracy convergence, then slow MI refinement) suggests agents first establish a functional protocol and then gradually synchronise their internal representations. Together with Experiment B1 (causal ZK taxonomy), C3 provides the informational complement: ZK measures whether communication is causally bidirectional; C3 measures whether it is informationally integrative.

Cross-Experiment Discussion

Twenty-three experiments now span two distinct but connected research programmes: emergent communication under channel constraints (Exp 1–3, 1b, 3b, 9b, 14, B1, B2, B3, C1, C3) and the structure of natural language as a window onto a universal concept space (Exp 4–15, A1, A2, A3, C2).

Message length is the dominant constraint; the optimal configuration is interior, not extreme. Experiment B2's full factorial grid search provides the definitive answer: when τ, k, L, and production cost are all varied simultaneously, message length produces the largest marginal effect on accuracy-weighted symbol quality (gap of 0.156 between $L = 4$ and $L = 1$, vs 0.043 for k and 0.025 for τ). The Pareto-efficient configuration ($\tau = 1.0$, $k = 32$, $L = 4$, cost$= 0.2$) has no setting at an extreme, confirming the non-monotonic opacity finding from Experiments 3 and 3b at a much larger hyperparameter grid.

Shannon's theorem is the unifying theoretical limit. Experiment C1 confirms that no protocol dominates on both compression and resilience. Experiment A3 extends this to multi-hop relay: Raw LaBSE degrades slowly; dense protocols degrade smoothly; text degrades step-wise after 2–3 hops. The relay behaviour at 200 concepts is consistent with the C1 Pareto structure.

The Phase 1 projection is cheap — exactly 3–5 pairs cheap. Experiment B3 quantifies what "cheap supervision" means: 3 antonym pairs achieve 50% of the maximum MRR gain; 9 pairs achieve 85%; 15 pairs saturate. The λ weight is irrelevant. For the A1-scale vocabulary, 50–100 WordNet antonym pairs should replicate the qualitative finding.

The ZK metric is a diagnostic taxonomy, not just a binary detector. Experiment B1 confirms that different failure modes produce distinct (CIC_1, RC, CIC_2) signatures. The one-way signalling signature (Mode 1: high CIC_2, near-zero RC) is distinguishable from full communication despite identical task accuracy. The additive ZK formula from Experiment 14 is further supported here.

Communication creates entanglement. Experiment C3 directly confirms the theoretical proposal's prediction: MI between two agents' separate world models rises monotonically during training (+0.045 nats vs −0.166 for severed baseline) and continues rising after task accuracy saturates.

Transitivity is NOT_DETECTED at definitive scale. Experiment A2 confirms NOT_DETECTED at 200 verbs (CV accuracy = majority baseline 0.745). The universal language requires explicit grammatical encoding of argument structure.

The UG findings scale, but category ARI is scale-sensitive. Zipf structure strengthens with scale (R^2: 0.708→0.861 raw, 0.855→0.915 projected). Frame role geometry persists ($\rho =$ 0.59/0.60, $p < 0.0001$). Category ARI drops from 0.519 to 0.137 — a metric artefact.

The language-design loss bottoms out at coverage, not expressivity. Experiment C2 finds that even 10 proto-tokens achieve full concept coverage. The binding constraint is the simplicity penalty. Tightening the coverage threshold to cosine $\geq$ 0.6–0.8 would reveal the actual vocabulary-size trade-off.

English occupies an unusual position and the bias is structural. Across Experiments 6, 7, 9, 10, 15, A1, and A2, English is consistently the most central language. Experiment 15 provides the most quantitatively precise demonstration: phoneme edit distance of 0.163 vs 0.789–0.921 for all other languages. Any production deployment must correct for this bias.

Abstract social concepts are the consistent failure mode. Experiments 4, 9 (Phase 3), 13, A1, and 15 all converge: concrete nouns, primary emotions, and physical-world concepts can be handled by centroid geometry alone; abstract social and aesthetic concepts require denser contextual annotation, more languages, or multimodal grounding.

The pipeline is complete end-to-end. The project has assembled every major component: a 4863-concept vocabulary with translatability tiers (A1); compositional projection requiring only 3–5 pairs (B3); frame role morphology confirmed at 409 concepts (A2); WALS-compliant phoneme inventory (15); dense AI-to-AI channel at $k = 4$ (8); ZK diagnostic taxonomy (B1); optimal constraint configuration $\tau = 1.0$, $k = 32$, $L = 4$, cost$= 0.2$ (B2); Shannon Pareto frontier (C1); multi-hop relay characterised at 200 concepts (A3); language-design loss operationalised (C2); and MI entanglement confirmed (C3). The next binding constraint is typological diversity and integration: running the full pipeline end-to-end at A1 scale with B2-optimal settings.

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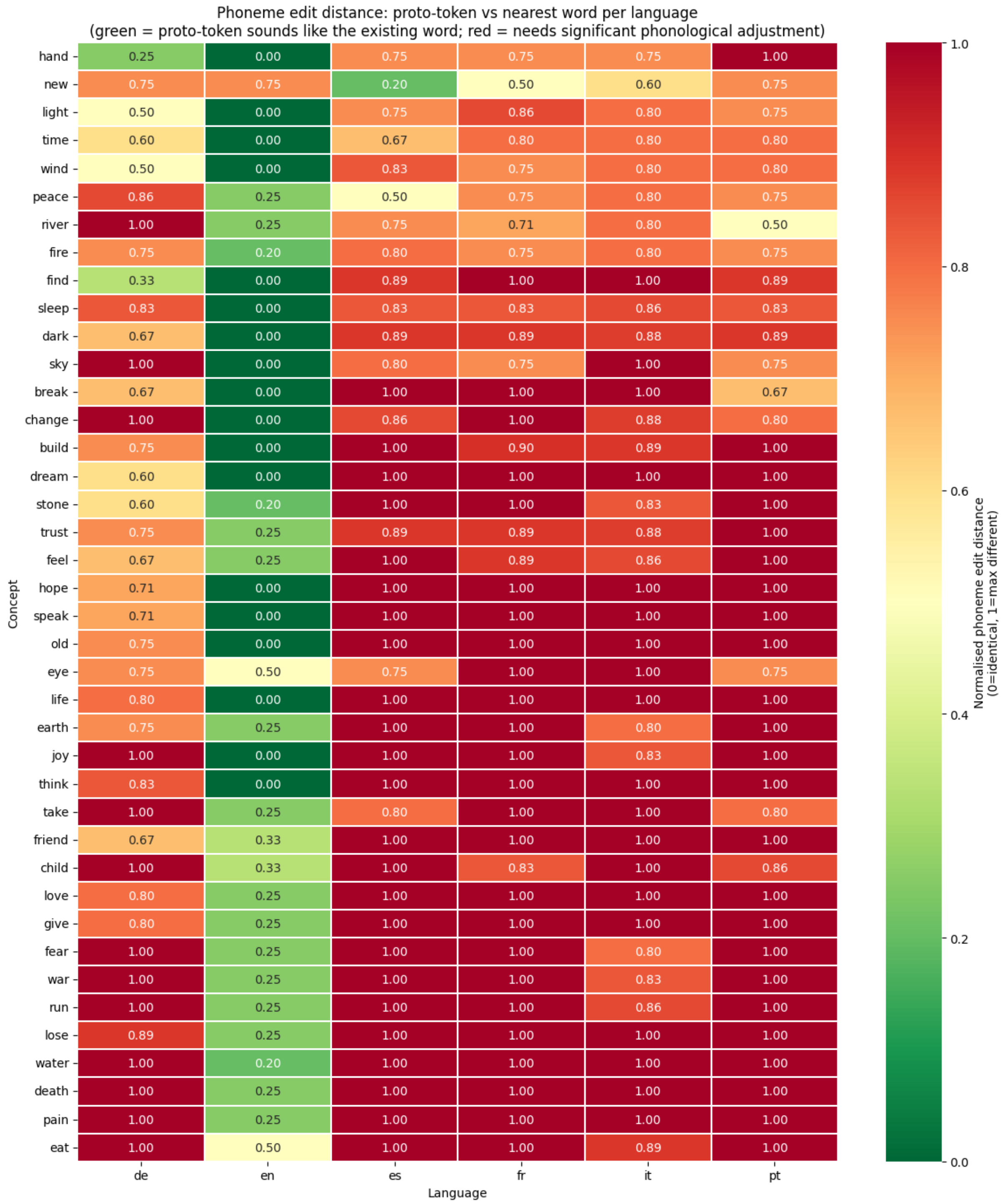


Figure 17: Phoneme edit distance heatmap (concept × language). German and English show the most green cells; French, Italian, and Portuguese are mostly red.

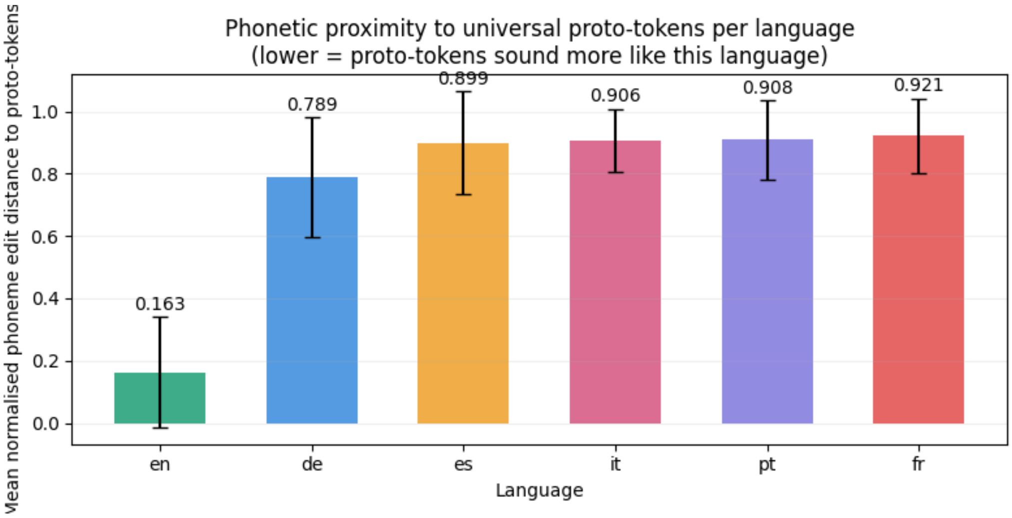


Figure 18: Mean normalised phoneme edit distance per language. English (0.163) is dramatically closer than all others (0.789–0.921).

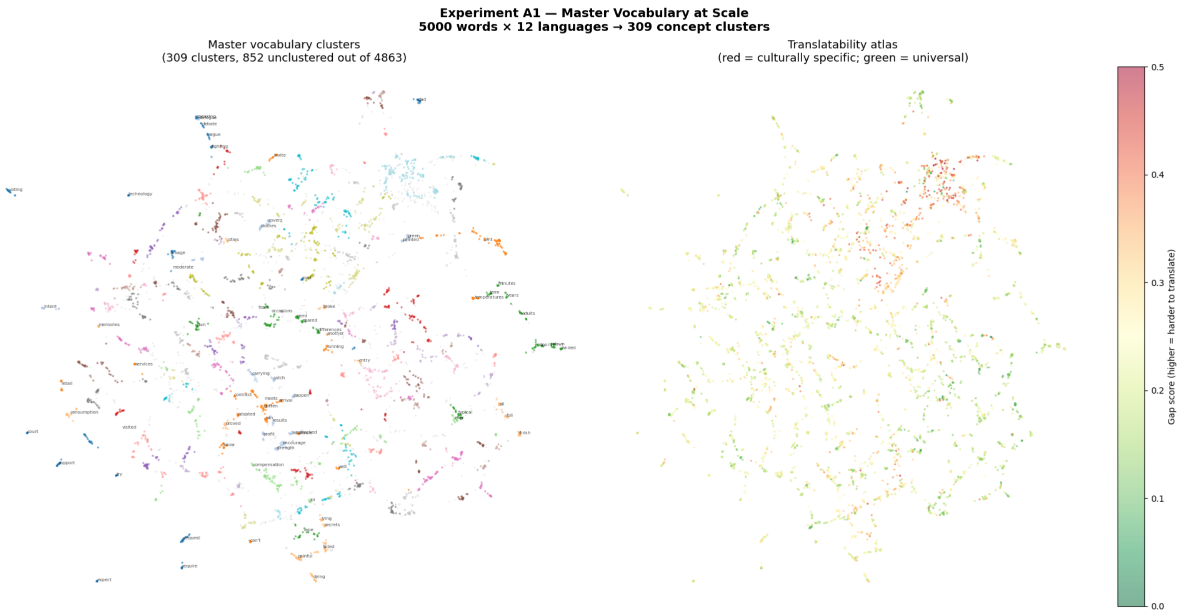


Figure 19: Experiment A1. *Left*: UMAP of 4,863 concept centroids coloured by cluster. *Right*: translatability atlas, coloured by gap score (green = universal, red = untranslatable). The universality gradient is geographically structured: the dense centre is predominantly green; the periphery is predominantly red.

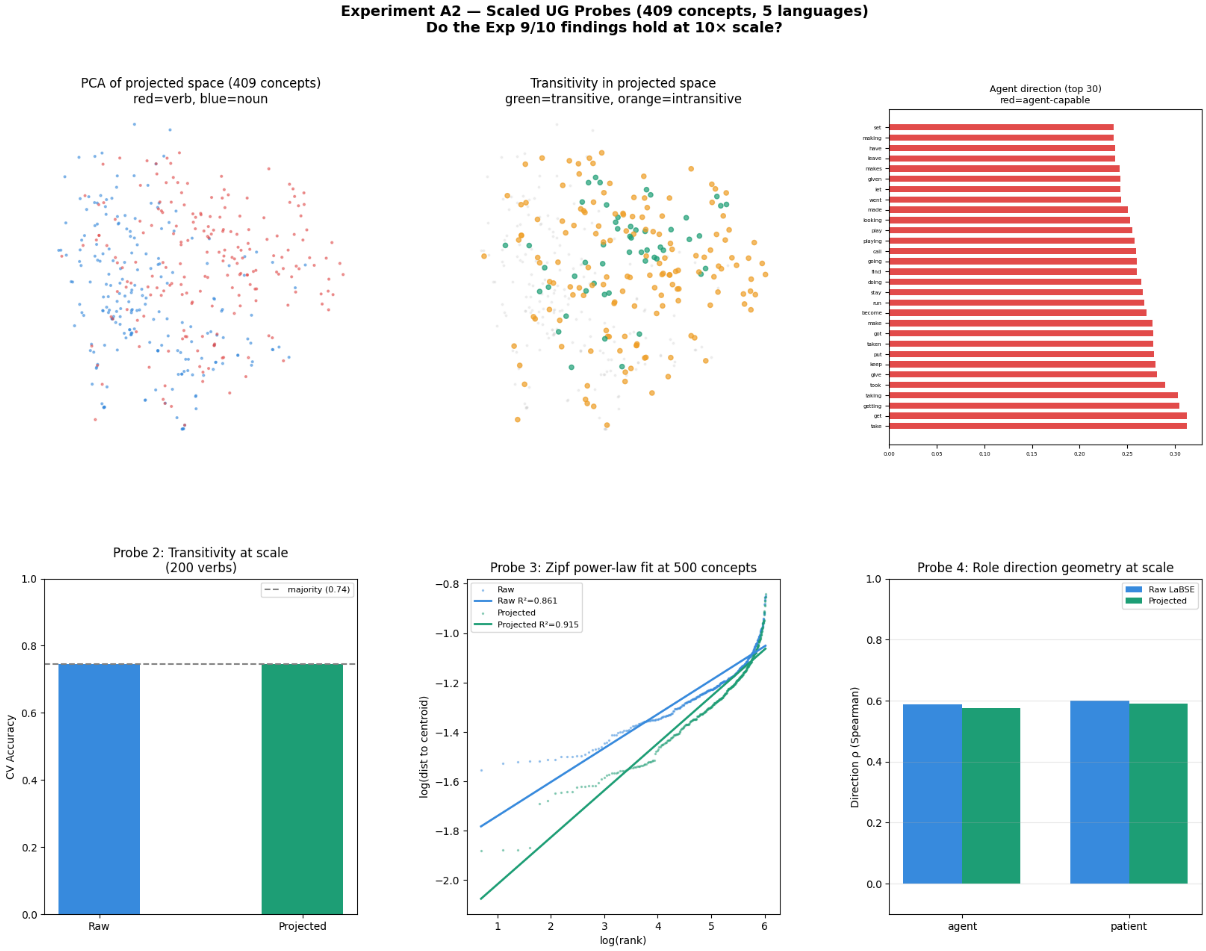


Figure 20: Experiment A2. *Top row*: PCA (verb/noun), transitivity in projected space (no spatial separation), agent direction bar chart (correct top-30 agent verbs). *Bottom row*: Probe 2 transitivity at majority baseline; Probe 3 Zipf fit; Probe 4 role direction ρ.

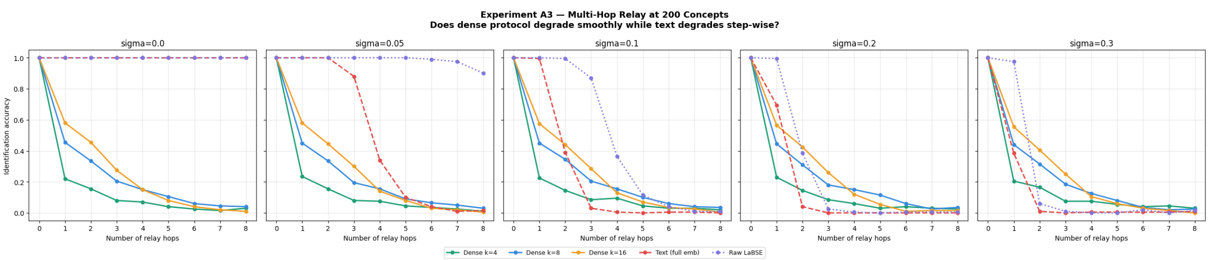
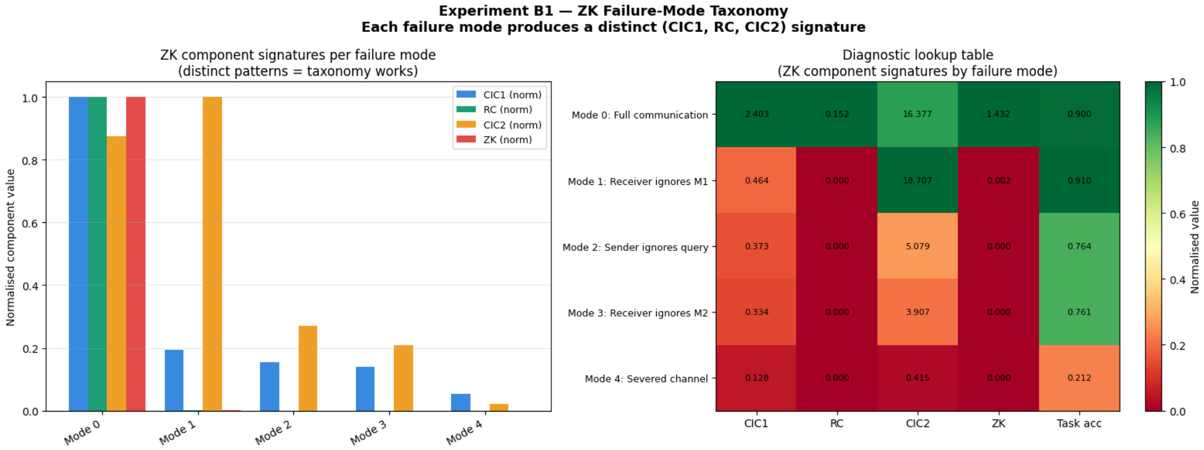


Figure 21: Experiment A3. Identification accuracy vs relay hops for five protocols at five noise levels. Raw LaBSE maintains near-perfect accuracy for 1–2 hops before collapsing. Text collapses sharply after 2 hops. Dense protocols degrade more gradually, with larger k more resilient.



	CIC1	RC	CIC2	ZK	Task acc
Mode 0: Full communication	2.403	0.152	16.377	1.432	0.900
Mode 1: Receiver ignores M1	0.464	0.000	18.707	0.002	0.910
Mode 2: Sender ignores query	0.373	0.000	5.079	0.000	0.764
Mode 3: Receiver ignores M2	0.334	0.000	3.907	0.000	0.761
Mode 4: Severed channel	0.128	0.000	0.415	0.000	0.212

Figure 22: Experiment B1. *Left*: normalised ZK components per failure mode. Mode 0 is the only condition with all three components elevated. *Right*: diagnostic lookup table heatmap.

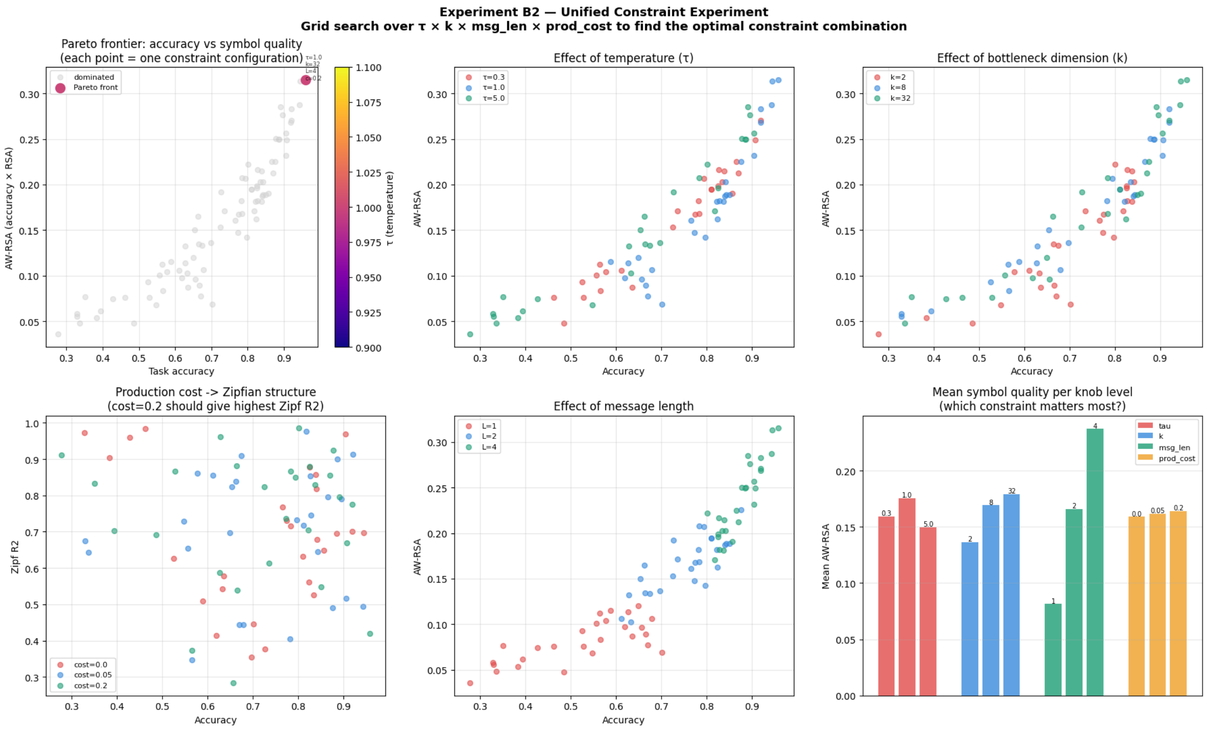


Figure 23: Experiment B2. Pareto frontier (top left); effects of τ, k, and L; production cost vs Zipf R^2; mean AW-RSA by knob level (bottom right).

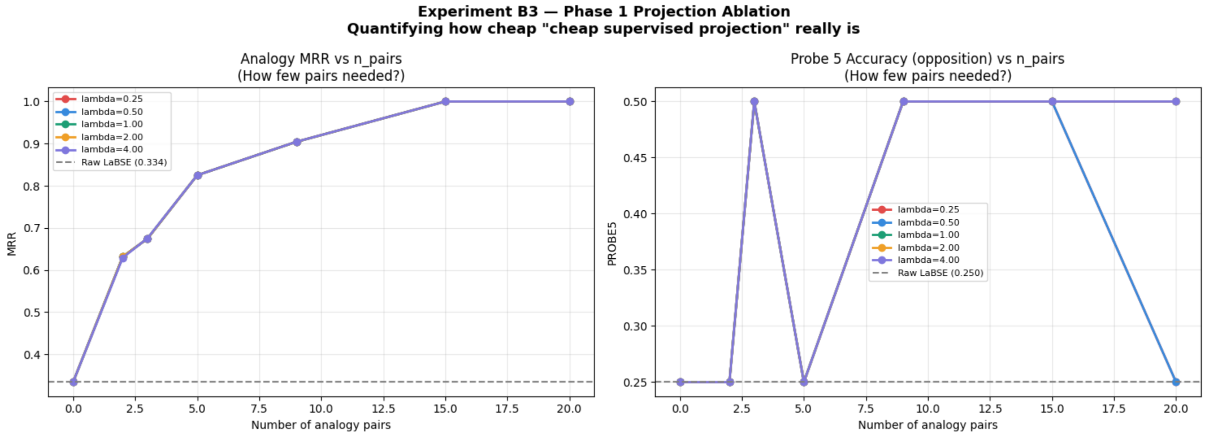


Figure 24: Experiment B3. *Left*: analogy MRR vs n_{pairs}. *Right*: Probe 5 opposition accuracy. All λ values produce nearly identical curves.

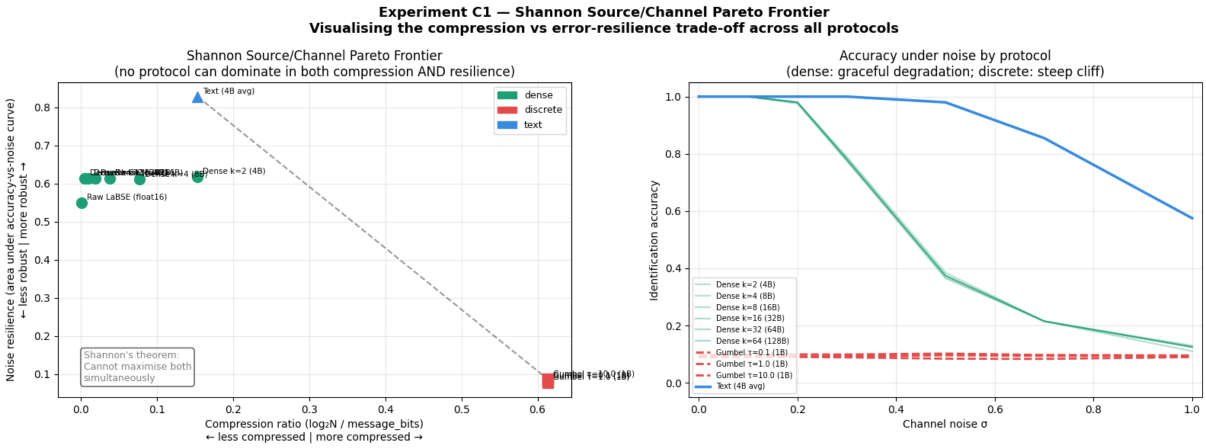


Figure 25: Experiment C1. *Left*: compression–resilience Pareto plane. *Right*: accuracy under noise by protocol family.

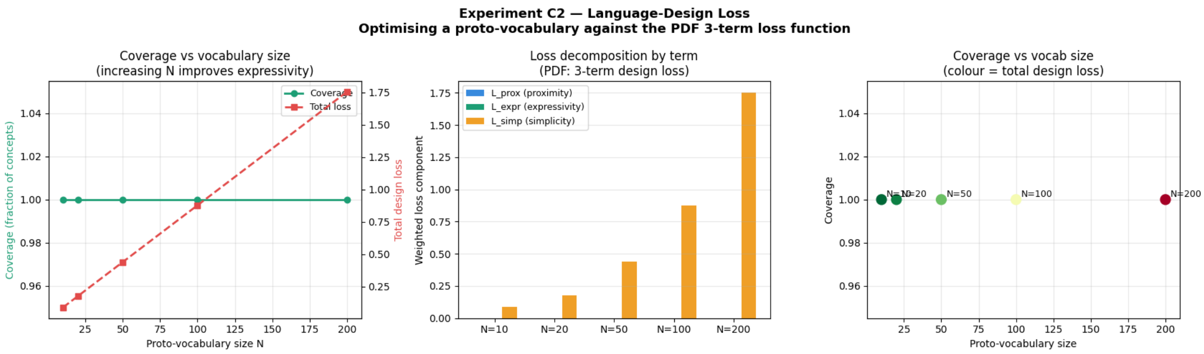


Figure 26: Experiment C2. Coverage and total loss vs N (left); loss decomposition (centre); coverage–N scatter (right).

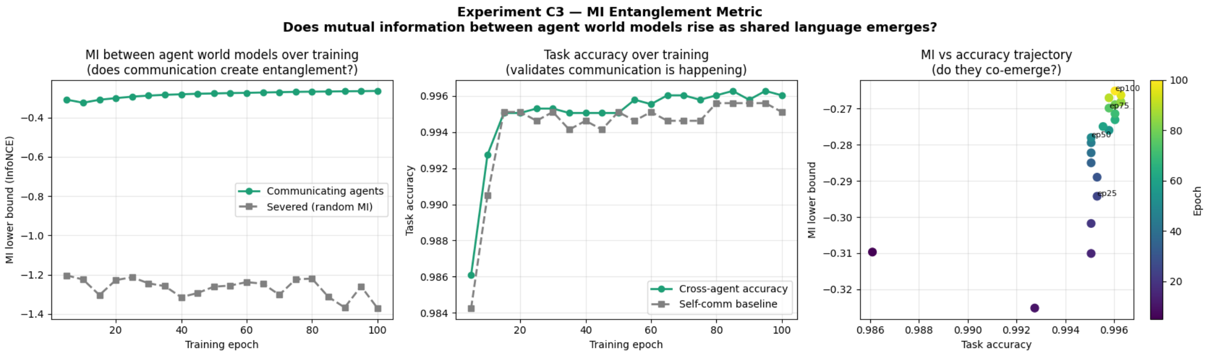


Figure 27: Experiment C3. *Left*: MI lower bound over training — communicating agents (green) rise; severed baseline (grey) flat. *Centre*: task accuracy. *Right*: MI vs accuracy trajectory.